GRUNDY DOMINATION OF GRAPH PRODUCTS AND MINIMUM RANK

DEEP BHATTACHARJEE

Abstract. The Grundy domination number of a strong product is at most the product of the minimum ranks of the factors over any field. We determine it for strong products of cycles, paths, complete bipartite graphs and Kneser graphs $K(n,2)$, proving in particular that $\gamma_{\mathrm{gr}}(C_k \boxtimes C_\ell) = (k-2)(\ell-2)$, and we obtain the analogous formula for Grundy total domination of direct products. Both product conjectures of Brešar et al. are false, with unbounded ratios, and incidence graphs of projective planes show that bipartite graphs without 4-cycles can have Grundy total domination number $o(n)$.

1. Introduction

1.1. **Dominating sequences.** All graphs are finite and simple. For a vertex v of a graph G we write $N(v)$ for its open and $N[v] = N(v) \cup \{v\}$ for its closed neighbourhood. A sequence $(v_1, \dots, v_k)$ of vertices is a *dominating sequence* if

$$N[v_i] \setminus \bigcup_{j<i} N[v_j] \neq \emptyset \qquad (1 \le i \le k), \tag{1.1}$$

and a *total dominating sequence* if the same holds with open neighbourhoods,

$$N(v_i) \setminus \bigcup_{j<i} N(v_j) \neq \emptyset \qquad (1 \le i \le k). \tag{1.2}$$

A vertex in the set on the left is a *footprint* of v_i.[1] The *Grundy domination number* $\gamma_{\mathrm{gr}}(G)$ is the largest length of a dominating sequence, and the *Grundy total domination number* $\gamma^t_{\mathrm{gr}}(G)$ is the largest length of a total dominating sequence; the second is defined for graphs without isolated vertices. These invariants were introduced by Brešar, Gologranc, Milanič, Rall and Rizzi [9] and by Brešar, Henning and Rall [10]. A longest sequence of either kind is a dominating, respectively total dominating, set, since an undominated vertex could be appended. So $\gamma_{\mathrm{gr}}(G)$ and $\gamma^t_{\mathrm{gr}}(G)$ are the largest sizes of such sets that can be built greedily while every new vertex enlarges the dominated set; they are the upper ends of the scale on which the domination number $\gamma(G)$ is the lower end. Computing either invariant is NP-hard [9, 10].

The two invariants are tied to zero forcing and to minimum rank. Brešar, Bujtás, Gologranc, Klavžar, Košmrlj, Patkós, Tuza and Vizer [8] showed that the zero forcing number of G is $|V(G)|$ minus a Grundy-type invariant of the same family, and Lin [13] showed that each of the Grundy invariants is bounded above by the minimum rank of a family of real symmetric matrices attached to the graph. The present paper turns this upper bound into a tool for graph products.

2020 *Mathematics Subject Classification.* Primary 05C69, 05C76; Secondary 05C50, 15A03, 51E15.

Key words and phrases. Grundy domination; graph products; minimum rank.

[1] The vertices of either kind of sequence are distinct, because a repeated vertex has no footprint. A vertex may have several footprints, and any one of them certifies the step.

1.2. **Products and the two conjectures.** In the strong product $G \boxtimes H$, two distinct vertices (g,h) and (g',h') are adjacent when $g' \in N[g]$ and $h' \in N[h]$, so that

$$N_{G\boxtimes H}[(g,h)] = N_G[g] \times N_H[h].$$

In the direct product $G \times H$ they are adjacent when $g' \in N(g)$ and $h' \in N(h)$, so that $N_{G\times H}((g,h)) = N_G(g) \times N_H(h)$. Listing the pairs of two longest sequences of the factors in lexicographic order gives

$$\gamma_{\rm gr}(G \boxtimes H) \ge \gamma_{\rm gr}(G)\,\gamma_{\rm gr}(H), \qquad \gamma^t_{\rm gr}(G \times H) \ge \gamma^t_{\rm gr}(G)\,\gamma^t_{\rm gr}(H), \tag{1.3}$$

for all graphs G and H (Lemma 2.3). Brešar et al. conjectured that both inequalities are equalities.

Conjecture 1.1 ([7, Conjecture 22])**.** *For all graphs G and H, $\gamma_{\rm gr}(G \boxtimes H) = \gamma_{\rm gr}(G)\,\gamma_{\rm gr}(H)$.*

Conjecture 1.2 ([6, Conjecture 2.2])**.** *For all graphs G and H without isolated vertices, $\gamma^t_{\rm gr}(G\times H) = \gamma^t_{\rm gr}(G)\,\gamma^t_{\rm gr}(H)$.*

Conjecture 1.1 was proved when one factor is a caterpillar [7] and, by Bell, Driscoll, Krop and Wolff [2], when one factor is a forest.[2] The exact value of $\gamma_{\rm gr}(C_k \boxtimes C_\ell)$, the Grundy domination number of the strong torus, was left open in [7], where the bound $\gamma_{\rm gr}(C_k \boxtimes C_\ell) \le (k-2)(\ell-1)$ was proved and the value $(k-2)(\ell-2)$ predicted by Conjecture 1.1 was stated as conjectural. Conjecture 1.2 was proved in [6] when one factor is a tree, when both factors are cycles, and in a few other cases, and the authors noted that it is equivalent to Conjecture 1.1 through a hypergraph formulation.

1.3. **Results.** For a field $\mathbb{F}$ and a graph G let $\mathrm{mr}_{\mathbb{F}}(G)$ be the least rank of a square matrix M over $\mathbb{F}$, with rows and columns indexed by $V(G)$, whose nonzero entries are exactly the pairs (u,v) with $v \in N[u]$; and let $\mathrm{mr}^\circ_{\mathbb{F}}(G)$ be the least rank of such a matrix whose nonzero entries are exactly the pairs with $v \in N(u)$. The diagonal of the first kind of matrix is nonzero and that of the second is zero. Unlike the minimum rank of the theory of zero forcing, the matrices need not be symmetric, the field is arbitrary, and the diagonal is prescribed.

Theorem A (rank bound)**.** *For every field $\mathbb{F}$ and all graphs G and H,*

$$\gamma_{\rm gr}(G \boxtimes H) \le \mathrm{mr}_{\mathbb{F}}(G)\ \mathrm{mr}_{\mathbb{F}}(H), \qquad \gamma^t_{\rm gr}(G \times H) \le \mathrm{mr}^\circ_{\mathbb{F}}(G)\ \mathrm{mr}^\circ_{\mathbb{F}}(H).$$

Theorem A rests on the fact that a dominating sequence produces a triangular submatrix with nonzero diagonal (Lemma 2.1), and that the Kronecker product of two matrices of the right shape has the shape of the product graph. It singles out the graphs for which the bound is attained. Call G *$\mathbb{F}$-tight* if $\gamma_{\rm gr}(G) = \mathrm{mr}_{\mathbb{F}}(G)$, and *totally $\mathbb{F}$-tight* if $\gamma^t_{\rm gr}(G) = \mathrm{mr}^\circ_{\mathbb{F}}(G)$. Strong products of $\mathbb{F}$-tight graphs are $\mathbb{F}$-tight and satisfy Conjecture 1.1, and direct products of totally $\mathbb{F}$-tight graphs satisfy Conjecture 1.2 (Corollary 2.5). Section 3 finds many tight graphs. The cycles are among them, which settles the torus.

Theorem B (tori and Kneser graphs)**.** *Let $r \ge 1$ and $k_1, \dots, k_r \ge 3$. Then*

$$\gamma_{\rm gr}(C_{k_1} \boxtimes C_{k_2} \boxtimes \cdots \boxtimes C_{k_r}) = \prod_{i=1}^{r}(k_i - 2).$$

More generally, the Grundy domination number of a strong product of cycles, paths, complete graphs and complete bipartite graphs is the product of the Grundy domination numbers of the factors. For every $n \ge 6$ the Kneser graph $K(n,2)$ is tight over $\mathbb{F}_2$ if n is even and over $\mathbb{F}_p$ for every odd prime p dividing $n-4$ if n is odd; so, for instance, $\gamma_{\rm gr}(K(n,2) \boxtimes C_k) = (n-1)(k-2)$

[2] A proof for all graphs was posted as arXiv:2212.04565 and withdrawn by its authors in its second version.

whenever n is even and $k \geq 3$ is odd, and $\gamma_{\mathrm{gr}}(K(n,2) \boxtimes K(m,2)) = (n-1)(m-1)$ whenever $n, m \geq 6$ are even.

The value $\gamma_{\mathrm{gr}}(K(n,2)) = n-1$ for $n \geq 6$ is due to Brešar, Kos and Torres [12]; the rank argument of Proposition 3.8 gives a short proof of the upper bound. The direct-product analogue holds for totally tight graphs, which include all cycles, trees, complete graphs and complete bipartite graphs (Theorem 6.4); it extends the theorem of [6] for two cycles to any number of cycles, trees and complete graphs.

Tightness is not universal, and the failure of tightness is exactly what the conjectures miss. The next theorem shows that both conjectures fail and that no constant multiple of the predicted value is an upper bound.

Theorem C (the conjectures fail)**.** (i) *For every $C > 0$ there are graphs G, H with $\gamma_{\mathrm{gr}}(G \boxtimes H) > C\,\gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H)$ and graphs G', H' without isolated vertices with $\gamma^t_{\mathrm{gr}}(G' \times H') > C\,\gamma^t_{\mathrm{gr}}(G')\,\gamma^t_{\mathrm{gr}}(H')$.*

(ii) *Let H_N be the graph on $\{0, 1, \dots, N-1\}$ in which u and v are adjacent when $|u - v| \equiv 1 \pmod 3$. For $N \geq 4$, $\gamma_{\mathrm{gr}}(H_N) = \lfloor N/3 \rfloor + 2$, while $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \geq N$; so Conjecture 1.1 fails for the octahedron $K_{2,2,2}$ and H_N when $N \in \{11, 13, 14\}$ or $N \geq 15$.*

The octahedron is $\mathbb{F}$-tight over every field with at least five elements (Proposition 3.5). So in part (ii) the graph H_N cannot be tight over any field, and indeed $\gamma_{\mathrm{gr}}(H_N) < \mathrm{mr}_{\mathbb{F}}(H_N)$ for every $\mathbb{F}$ and every N in the range of (ii) (Corollary 5.7). For the direct product, the passage from closed to open neighbourhoods is a doubling construction: if $\mathsf{B}(G)$ is the bipartite graph on $V(G) \times \{0,1\}$ in which $(u,0)$ is adjacent to $(v,1)$ when $v \in N[u]$, then $\gamma^t_{\mathrm{gr}}(\mathsf{B}(G)) = 2\gamma_{\mathrm{gr}}(G)$ and $\mathsf{B}(G) \times \mathsf{B}(H)$ is two disjoint copies of $\mathsf{B}(G \boxtimes H)$ (Proposition 6.5). The smallest counterexample to Conjecture 1.1 that we know has 66 vertices, and the smallest one to Conjecture 1.2 has 264.

The last theorem concerns a question of Brešar, Henning and Rall [10, Question 1]. They showed that there is no constant $c > 0$ with $\gamma^t_{\mathrm{gr}}(G) \geq c|V(G)|$ for all bipartite graphs without open twins, noticed that every vertex of their examples lies on a 4-cycle, and asked whether such a constant exists for bipartite graphs without 4-cycles and with minimum degree at least 2.

Theorem D (projective planes)**.** *Let $q = p^e$ be a prime power and let $\mathrm{Inc}(q)$ be the point–line incidence graph of the projective plane $\mathrm{PG}(2,q)$. It is bipartite, $(q+1)$-regular, has girth 6 and $n = 2(q^2+q+1)$ vertices, and*

$$\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(q)) \leq 2\Big(\tbinom{p+1}{2}^e + 1\Big).$$

In particular $\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(2^e)) \leq 2(3^e+1)$, and $\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(p^e))/|V(\mathrm{Inc}(p^e))| \to 0$ as $e \to \infty$ for every fixed prime p. So the answer to the question of Brešar, Henning and Rall is negative, even for regular graphs of girth 6 and arbitrarily large degree. For $q \in \{2,3,4,8\}$ the bound is attained.

1.4. **How the paper is organised.** Section 2 proves the rank lemma and Theorem A. Section 3 constructs tight graphs: complete, complete bipartite and cocktail party graphs, cycles, paths and Kneser graphs, and Section 4 assembles them into Theorem B and a table of exact values. Section 5 proves the strong-product part of Theorem C, first with two graphs on the five-element subsets of an m-set, whose factor bounds come from incidence matrices over two different fields, and then with the octahedron and the graphs H_N. Section 6 treats direct products and Grundy total domination and proves the remaining part of Theorem C. Section 7 proves Theorem D. Section 8 reports exact computations and a search for small counterexamples, and Section 9 lists the open problems, with a table of the status of every conjecture and question treated here (Table 4). The appendices contain explicit sequences and matrices and describe the independent machine checks.

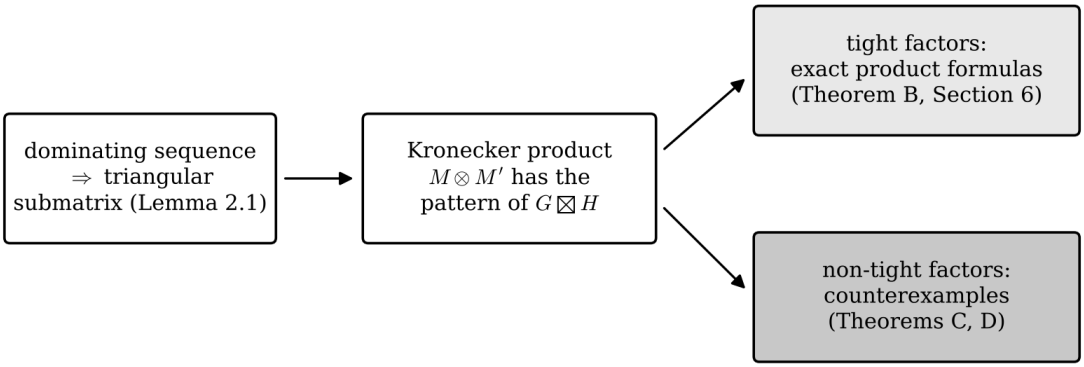


FIGURE 1. The structure of the paper. A dominating sequence gives a triangular submatrix (left), the Kronecker product carries matrices to products (centre), and tight factors give exact product formulas, while non-tight factors give the counterexamples (right).

The method is linear algebra over finite fields together with elementary counting. Counting arguments attached to a finite structure or to a group action appear also in the author's earlier work [3, 4, 5].

2. Triangular submatrices and the rank bound

2.1. **Pattern matrices.** Let G be a graph and $\mathbb{F}$ a field. A square matrix M over $\mathbb{F}$ with rows and columns indexed by $V(G)$ is a *closed pattern matrix* of G if

$$M_{uv} \neq 0 \iff v \in N[u] \qquad (u, v \in V(G)),$$

and an *open pattern matrix* of G if $M_{uv} \neq 0 \iff v \in N(u)$. The two sets of matrices are denoted $\mathcal{S}_{\mathbb{F}}[G]$ and $\mathcal{S}_{\mathbb{F}}(G)$, and

$$\mathrm{mr}_{\mathbb{F}}(G) = \min\{\operatorname{rank} M : M \in \mathcal{S}_{\mathbb{F}}[G]\}, \qquad \mathrm{mr}^{\circ}_{\mathbb{F}}(G) = \min\{\operatorname{rank} M : M \in \mathcal{S}_{\mathbb{F}}(G)\}.$$

The matrices are not required to be symmetric. If G has no isolated vertex, then $\mathcal{S}_{\mathbb{F}}(G)$ is nonempty; $\mathcal{S}_{\mathbb{F}}[G]$ always contains $A + I$, where A is the adjacency matrix, and the identity of $\mathbb{F}$ is used for the nonzero entries.

Two elementary remarks will be used repeatedly. First, if $\mathbb{F} \subseteq \mathbb{F}'$ is a field extension, then a matrix over $\mathbb{F}$ has the same rank over $\mathbb{F}'$, so

$$\mathrm{mr}_{\mathbb{F}'}(G) \leq \mathrm{mr}_{\mathbb{F}}(G), \qquad \mathrm{mr}^{\circ}_{\mathbb{F}'}(G) \leq \mathrm{mr}^{\circ}_{\mathbb{F}}(G). \tag{2.1}$$

Second, both invariants are additive over disjoint unions, by block-diagonal matrices, and the Grundy invariants are additive as well, because a sequence in a disjoint union is legal exactly when its restriction to each component is legal.[3]

2.2. **The rank lemma.**

Lemma 2.1. *Let G be a graph and $\mathbb{F}$ a field. Then $\gamma_{\mathrm{gr}}(G) \leq \mathrm{mr}_{\mathbb{F}}(G)$, and if G has no isolated vertex, $\gamma^{t}_{\mathrm{gr}}(G) \leq \mathrm{mr}^{\circ}_{\mathbb{F}}(G)$.*

[3]The domination steps in different components do not interact: $N[v]$ lies in the component of v.

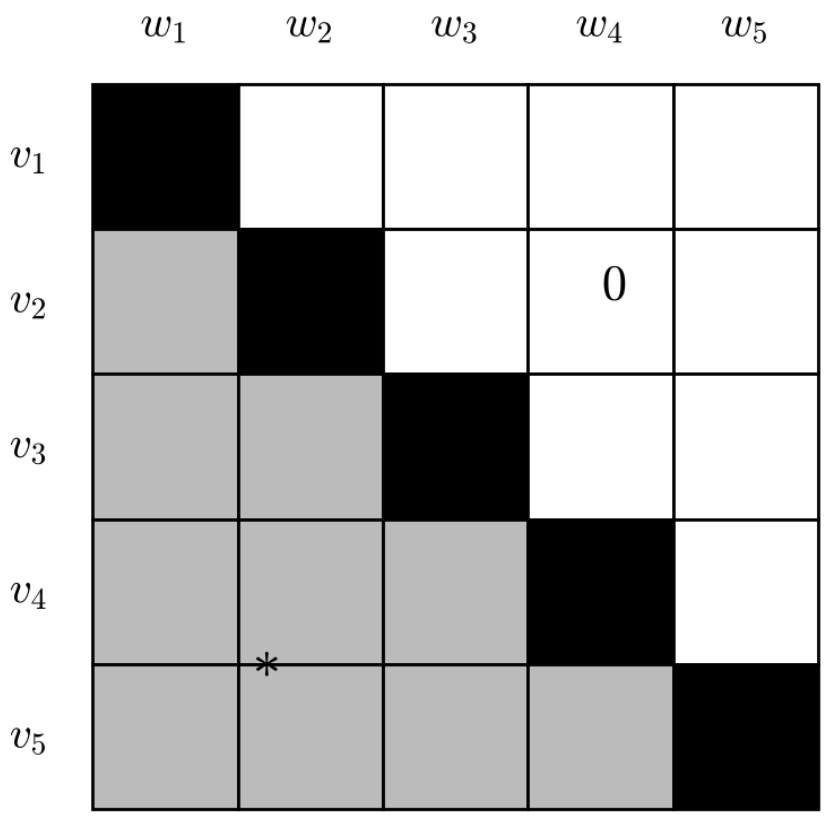


FIGURE 2. A dominating sequence $(v_1, \dots, v_5)$ with footprints $w_1, \dots, w_5$ selects a lower triangular submatrix with nonzero diagonal from any pattern matrix. The black cells are the diagonal, the white cells above it vanish, and the grey cells below it are unconstrained.

Proof. Let $(v_1, \dots, v_k)$ be a dominating sequence and choose a footprint w_i of v_i for each i. If $j < i$, then $w_i \notin N[v_j]$, while $w_j \in N[v_j]$; hence $w_i \neq w_j$, and the footprints are distinct. Let $M \in \mathcal{S}_{\mathbb{F}}[G]$ and consider the $k \times k$ submatrix T with rows $v_1, \dots, v_k$ and columns $w_1, \dots, w_k$, in this order. Its diagonal entries $M_{v_i w_i}$ are nonzero since $w_i \in N[v_i]$, and its entries above the diagonal vanish, because for $j < i$ the entry in row v_j and column w_i is $M_{v_j w_i} = 0$ as $w_i \notin N[v_j]$. So T is lower triangular with nonzero diagonal, $\operatorname{rank} T = k$, and $k \leq \operatorname{rank} M$. Taking a longest sequence and a matrix of least rank gives $\gamma_{\mathrm{gr}}(G) \leq \mathrm{mr}_{\mathbb{F}}(G)$. The total case is identical, with open neighbourhoods and $M \in \mathcal{S}_{\mathbb{F}}(G)$. □

For real symmetric matrices, and with the diagonal free in the appropriate way, the inequality is Lin's [13, Theorem 3.1]. The argument needs neither symmetry nor any particular field, but it does need the nonzero entries to be exactly the prescribed ones: a footprint w_i is in general a neighbour of v_i and not v_i itself, so the entries M_{uv} with $v \in N(u)$ must be nonzero, and the entries outside the neighbourhoods must vanish (Figure 2).

Remark 2.2. The argument also shows what $\gamma_{\mathrm{gr}}(G)$ is in matrix terms: it is the largest k for which there are distinct vertices $v_1, \dots, v_k$ and distinct vertices $w_1, \dots, w_k$ such that $w_i \in N[v_i]$ for all i and $w_i \notin N[v_j]$ for $j < i$. In other words, $\gamma_{\mathrm{gr}}(G)$ is the size of the largest lower triangular

submatrix with nonzero diagonal, after independent permutations of rows and columns, of the $0/1$ matrix $A+I$. The rank bound replaces this combinatorial quantity, which is not monotone under taking matrices with the same pattern, by a linear-algebraic one that is.

2.3. **Lower bounds for products.**

Lemma 2.3. *For all graphs G and H, $\gamma_{\mathrm{gr}}(G \boxtimes H) \geq \gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$; if G and H have no isolated vertices, $\gamma^t_{\mathrm{gr}}(G \times H) \geq \gamma^t_{\mathrm{gr}}(G)\gamma^t_{\mathrm{gr}}(H)$.*

Proof. Let $(g_1, \ldots, g_a)$ and $(h_1, \ldots, h_b)$ be dominating sequences of G and H with footprints x_i and y_j. List the pairs (g_i, h_j) in lexicographic order. The pair (x_i, y_j) lies in $N[g_i] \times N[h_j] = N[(g_i, h_j)]$. An earlier pair (g_s, h_t) has either $s < i$, and then $x_i \notin N[g_s]$, or $s = i$ and $t < j$, and then $y_j \notin N[h_t]$; in both cases $(x_i, y_j) \notin N[(g_s, h_t)]$. So (x_i, y_j) is a footprint of (g_i, h_j) and the sequence has length ab. The total case is the same with open neighbourhoods and the direct product. □

The first inequality is [7, Proposition 21] and the second is [6, Lemma 2.1]. The same proof works for any number of factors: for graphs $G_1, \ldots, G_r$,

$$\gamma_{\mathrm{gr}}(G_1 \boxtimes \cdots \boxtimes G_r) \geq \prod_{i=1}^{r} \gamma_{\mathrm{gr}}(G_i), \qquad \gamma^t_{\mathrm{gr}}(G_1 \times \cdots \times G_r) \geq \prod_{i=1}^{r} \gamma^t_{\mathrm{gr}}(G_i). \tag{2.2}$$

2.4. **The rank bound for products.**

Proof of Theorem A. Let $M \in \mathcal{S}_{\mathbb{F}}[G]$ and $M' \in \mathcal{S}_{\mathbb{F}}[H]$ have ranks $\mathrm{mr}_{\mathbb{F}}(G)$ and $\mathrm{mr}_{\mathbb{F}}(H)$. The Kronecker product $M \otimes M'$ has rows and columns indexed by $V(G) \times V(H)$, and its entry in row (g, h) and column (g', h') is $M_{gg'}M'_{hh'}$. In a field a product is nonzero exactly when both factors are, so this entry is nonzero exactly when $g' \in N[g]$ and $h' \in N[h]$, that is, when $(g', h') \in N[(g, h)]$ in $G \boxtimes H$. Thus $M \otimes M' \in \mathcal{S}_{\mathbb{F}}[G \boxtimes H]$, and since $\mathrm{rank}(M \otimes M') = \mathrm{rank}\, M \cdot \mathrm{rank}\, M'$, Lemma 2.1 gives the first bound. For the second, if $M \in \mathcal{S}_{\mathbb{F}}(G)$ and $M' \in \mathcal{S}_{\mathbb{F}}(H)$, then $M_{gg'}M'_{hh'} \neq 0$ exactly when $g' \in N(g)$ and $h' \in N(h)$, which is adjacency in $G \times H$; so $M \otimes M' \in \mathcal{S}_{\mathbb{F}}(G \times H)$. □

By induction, $\gamma_{\mathrm{gr}}(G_1 \boxtimes \cdots \boxtimes G_r) \leq \prod_i \mathrm{mr}_{\mathbb{F}}(G_i)$ and the same for direct products, and in fact

$$\mathrm{mr}_{\mathbb{F}}(G \boxtimes H) \leq \mathrm{mr}_{\mathbb{F}}(G)\ \mathrm{mr}_{\mathbb{F}}(H), \qquad \mathrm{mr}^{\circ}_{\mathbb{F}}(G \times H) \leq \mathrm{mr}^{\circ}_{\mathbb{F}}(G)\ \mathrm{mr}^{\circ}_{\mathbb{F}}(H). \tag{2.3}$$

Definition 2.4. A graph G is $\mathbb{F}$-*tight* if $\gamma_{\mathrm{gr}}(G) = \mathrm{mr}_{\mathbb{F}}(G)$, and *totally* $\mathbb{F}$-*tight* if it has no isolated vertices and $\gamma^t_{\mathrm{gr}}(G) = \mathrm{mr}^{\circ}_{\mathbb{F}}(G)$. It is *tight* if it is $\mathbb{F}$-tight for some field $\mathbb{F}$.

By (2.1), an $\mathbb{F}$-tight graph is $\mathbb{F}'$-tight for every extension $\mathbb{F}'$ of $\mathbb{F}$: the minimum rank can only go down under extension, and it cannot go below $\gamma_{\mathrm{gr}}(G)$. So if G is $\mathbb{F}$-tight and H is $\mathbb{F}'$-tight, and $\mathbb{F}$ and $\mathbb{F}'$ have the same characteristic, then both are tight over a common field, a field containing copies of both.[4]

Corollary 2.5. *If $G_1, \ldots, G_r$ are $\mathbb{F}$-tight, then*

$$\gamma_{\mathrm{gr}}(G_1 \boxtimes \cdots \boxtimes G_r) = \prod_{i=1}^{r} \gamma_{\mathrm{gr}}(G_i),$$

and the product is $\mathbb{F}$-tight. If $G_1, \ldots, G_r$ are totally $\mathbb{F}$-tight, then $\gamma^t_{\mathrm{gr}}(G_1 \times \cdots \times G_r) = \prod_i \gamma^t_{\mathrm{gr}}(G_i)$ and the product is totally $\mathbb{F}$-tight. In particular, if G and H are tight over fields of the same characteristic, then G and H satisfy Conjecture 1.1.

[4]Two fields of the same characteristic embed in a common field, for instance in an algebraically closed field of that characteristic of sufficiently large transcendence degree. In all our examples a finite field or $\mathbb{C}$ suffices.

Proof. The lower bound is (2.2). The upper bound is Theorem A with (2.3), which also gives $\mathrm{mr}_{\mathbb{F}}(\prod G_i) \le \prod \gamma_{\mathrm{gr}}(G_i) = \gamma_{\mathrm{gr}}(\prod G_i)$; with Lemma 2.1 this is an equality, so the product is tight. □

A counterexample to Conjecture 1.1 therefore needs, for every characteristic, a factor that is not tight in that characteristic. The constructions of Section 5 show how this happens.

Remark 2.6. Two vertices $u \neq v$ are *true twins* if $N[u] = N[v]$. If v is a true twin of u, then $\gamma_{\mathrm{gr}}(G - v) = \gamma_{\mathrm{gr}}(G)$, since a dominating sequence can contain at most one of u, v (after one of them is chosen the closed neighbourhood of the other is dominated) and v can be replaced by u. Duplicating the row and column of u in a closed pattern matrix of $G - v$ gives one of G with the same rank. Hence G is $\mathbb{F}$-tight if and only if $G - v$ is. In particular every graph obtained from an $\mathbb{F}$-tight graph by substituting complete graphs for vertices is $\mathbb{F}$-tight; compare [7, Proposition 30].

Justification of the last claim in Remark 2.6. If M is a closed pattern matrix of $G - v$ and M' is obtained by repeating row u as row v and column u as column v, then $M'_{vv} = M'_{uv} = M'_{vu} = M_{uu} \neq 0$, which is correct since u, v are adjacent, and for $w \neq u, v$ the entries in row and column v copy those of u, which is correct since $N[v] = N[u]$. Repeating a row and a column does not change the rank. Conversely, deleting a row and a column cannot increase the rank, so $\mathrm{mr}_{\mathbb{F}}(G - v) \le \mathrm{mr}_{\mathbb{F}}(G)$, and the inequalities $\gamma_{\mathrm{gr}}(G) = \gamma_{\mathrm{gr}}(G - v) \le \mathrm{mr}_{\mathbb{F}}(G - v) \le \mathrm{mr}_{\mathbb{F}}(G)$ show that tightness passes in both directions. □

3. Tight graphs

This section produces the tight graphs used in the product formulas. In each case the lower bound for γ_{gr} is an explicit sequence and the upper bound is a matrix of the right pattern and rank.

3.1. **Complete graphs, edgeless graphs and induced subgraphs.**

The all-ones matrix J is a closed pattern matrix of K_n of rank 1, and $\gamma_{\mathrm{gr}}(K_n) = 1$; the identity is a closed pattern matrix of the edgeless graph $\overline{K_n}$ of rank $n = \gamma_{\mathrm{gr}}(\overline{K_n})$. So both are tight over every field. We record a general principle for lower bounds.

Lemma 3.1. *If F is an induced subgraph of G, then every dominating sequence of F is a dominating sequence of G with the same footprints. Hence $\gamma_{\mathrm{gr}}(F) \le \gamma_{\mathrm{gr}}(G)$.*

Proof. For $v \in V(F)$ we have $N_F[v] = N_G[v] \cap V(F)$, so a vertex of F that lies in $N_F[v_i]$ and in no $N_F[v_j]$, $j < i$, lies in $N_G[v_i]$ and in no $N_G[v_j]$. □

Example 3.2. The path P_m with vertices $1, 2, \dots, m$, $m \ge 2$, has the dominating sequence $(1, 2, \dots, m-1)$: the vertex i footprints $i + 1$, which no earlier vertex dominates. So $\gamma_{\mathrm{gr}}(P_m) \ge m - 1$. Since a dominating sequence of a graph without isolated vertices cannot use every vertex,[5] $\gamma_{\mathrm{gr}}(P_m) = m - 1$. By Lemma 3.1, a graph containing an induced path on m vertices has $\gamma_{\mathrm{gr}} \ge m - 1$; in particular $\gamma_{\mathrm{gr}}(C_k) \ge k - 2$.

3.2. **Paths.**

Proposition 3.3. *For $m \ge 2$ the path P_m is $\mathbb{R}$-tight.*

Proof. Let $Q = D + A$ be the signless Laplacian of P_m, where D is the diagonal matrix of degrees. Its diagonal entries are 1 or 2 and its off-diagonal entries are 1 exactly on the edges, so $Q \in \mathcal{S}_{\mathbb{R}}[P_m]$. Writing $Q = BB^{\mathsf{T}}$ with B the $m \times (m-1)$ vertex–edge incidence matrix, we have $\operatorname{rank} Q = \operatorname{rank} B \le m - 1$. With Example 3.2, $m - 1 = \gamma_{\mathrm{gr}}(P_m) \le \mathrm{mr}_{\mathbb{R}}(P_m) \le m - 1$. □

[5] The last vertex v_n of a sequence using all n vertices would need a footprint outside $N[v_1] \cup \cdots \cup N[v_{n-1}]$, which is the whole vertex set as soon as every vertex has a neighbour.

3.3. **Complete bipartite graphs and cocktail party graphs.**

Proposition 3.4. *Let $a \geq b \geq 1$ and let $\mathbb{F}$ be a field in which $a \neq 0$ and which contains a primitive a-th root of unity. Then $K_{a,b}$ is $\mathbb{F}$-tight and $\gamma_{\mathrm{gr}}(K_{a,b}) = a$. In particular $K_{a,b}$ is $\mathbb{C}$-tight.*

Proof. Listing the a vertices of the larger side gives a dominating sequence, each vertex footprinting itself, so $\gamma_{\mathrm{gr}}(K_{a,b}) \geq a$. Let $\zeta \in \mathbb{F}$ be a primitive a-th root of unity and define the $a \times b$ matrix Z and the $b \times a$ matrix W by $Z_{kj} = \zeta^{kj}$ and $W_{jk} = \zeta^{-kj}$, for $0 \leq k < a$ and $0 \leq j < b$. Then

$$(WZ)_{jj'} = \sum_{k=0}^{a-1} \zeta^{k(j'-j)} = a\,\delta_{jj'},$$

because for $0 < |j' - j| < a$ the element $\zeta^{j'-j} \neq 1$ is a root of $1 + x + \cdots + x^{a-1}$. Put

$$M = \begin{pmatrix} aI_a & Z \\ W & I_b \end{pmatrix}.$$

Its nonzero entries are the diagonal and all entries between the two sides, which is the closed pattern of $K_{a,b}$, and its last b rows are $a^{-1}W$ times its first a rows, since $a^{-1}W(aI_a\ Z) = (W\ I_b)$. So $\operatorname{rank} M = a$. □

The *cocktail party graph* $CP(s) = K_{2,2,\ldots,2}$, with $s \geq 2$ parts of size two, is the complete graph K_{2s} minus a perfect matching. For $s = 3$ it is the octahedron. Each vertex is adjacent to every other vertex except its *antipode*.

Proposition 3.5. *For $s \geq 2$, $\gamma_{\mathrm{gr}}(CP(s)) = 2$, and $CP(s)$ is $\mathbb{F}$-tight over every field $\mathbb{F}$ with at least $2s - 1$ elements.*

Proof. A vertex u followed by its antipode $\bar{u}$ is a dominating sequence: $\bar{u}$ footprints itself. Conversely, after two vertices u, v of a sequence everything is dominated: if $v = \bar{u}$ then $N[u] \cup N[v]$ is the whole vertex set, and otherwise $N[u]$ misses only $\bar{u}$, which lies in $N[v]$. So $\gamma_{\mathrm{gr}}(CP(s)) = 2$.

For the matrix, identify $\mathbb{F}^2$ with row vectors and let $\langle x, y\rangle = x_1y_1 + x_2y_2$. For $x \in \mathbb{F}^2$ write $x^\perp = (-x_2, x_1)$, so that $\langle y, x^\perp\rangle = 0$ exactly when y is a multiple of x. Since $\mathbb{F}$ has at least $2s-1$ elements, $\mathbb{F}^2$ has $|\mathbb{F}|+1 \geq 2s$ distinct one-dimensional subspaces; choose nonzero vectors r_u, one for each of the $2s$ vertices u, spanning pairwise distinct lines. Define

$$M_{uv} = \langle r_u,\, r_{\bar{v}}^{\perp}\rangle.$$

Then M has rank at most 2, since $M = RC$ with R the $2s \times 2$ matrix of the rows r_u and C the $2 \times 2s$ matrix with columns $r_{\bar{v}}^{\perp\mathsf{T}}$. The entry M_{uv} vanishes exactly when r_u is a multiple of $r_{\bar{v}}$, that is, when $u = \bar{v}$; so the zero entries of M are exactly the antipodal pairs, and the diagonal entries are nonzero because $u \neq \bar{u}$. Thus $M \in \mathcal{S}_{\mathbb{F}}[CP(s)]$ and $\mathrm{mr}_{\mathbb{F}}(CP(s)) \leq 2 = \gamma_{\mathrm{gr}}(CP(s))$. □

For the octahedron $CP(3)$ this needs $|\mathbb{F}| \geq 5$; it is tight over $\mathbb{R}$, over $\mathbb{C}$ and over every finite field with at least five elements.

3.4. **Cycles.**

Proposition 3.6. *Let $k \geq 3$, $k \neq 4$, and let $\mathbb{F}$ be a field whose characteristic does not divide k and which contains a primitive k-th root of unity ζ. Then C_k is $\mathbb{F}$-tight and $\gamma_{\mathrm{gr}}(C_k) = k - 2$. The same holds over $\mathbb{F} = \mathbb{R}$, and $C_4 = K_{2,2}$ is tight over every field of characteristic different from 2 that contains $-1 \neq 1$, in particular over $\mathbb{C}$. Hence every cycle is $\mathbb{C}$-tight.*

Proof. Label the vertices by $\mathbb{Z}/k$ with i adjacent to $i \pm 1$, let A be the adjacency matrix and put

$$d = -(\zeta + \zeta^{-1}), \qquad M = A + dI.$$

First, $d \neq 0$: otherwise $\zeta^2 = -1$ and $\zeta^4 = 1$, so the order k of ζ divides 4, which is excluded for $k \geq 3$, $k \neq 4$. So $M \in \mathcal{S}_{\mathbb{F}}[C_k]$. The matrix M is circulant. Let V be the $k \times k$ matrix $V_{jl} = \zeta^{jl}$; it is a Vandermonde matrix on the distinct elements $1, \zeta, \ldots, \zeta^{k-1}$, hence invertible. For the column $v_j = (\zeta^{jl})_l$ of V we have

$$(Mv_j)_l = \zeta^{j(l+1)} + \zeta^{j(l-1)} + d\,\zeta^{jl} = \lambda_j\,\zeta^{jl}, \qquad \lambda_j = \zeta^j + \zeta^{-j} - \zeta - \zeta^{-1},$$

so M is diagonalised by V with eigenvalues $\lambda_0, \ldots, \lambda_{k-1}$. For $x, y \neq 0$,

$$x + x^{-1} - y - y^{-1} = (x - y)(1 - (xy)^{-1}),$$

so $\lambda_j = 0$ exactly when $\zeta^j = \zeta$ or $\zeta^j = \zeta^{-1}$, that is, when $j \equiv \pm 1 \pmod{k}$. These are two values of j since $k \geq 3$, and so $\operatorname{rank} M = k - 2$. With Example 3.2, $k - 2 \leq \gamma_{\mathrm{gr}}(C_k) \leq \mathrm{mr}_{\mathbb{F}}(C_k) \leq k - 2$.

Over $\mathbb{R}$ the same computation, with $d = -2\cos(2\pi/k)$ and the real circulant $M = A + dI$, gives the eigenvalues $2\cos(2\pi j/k) - 2\cos(2\pi/k)$, which vanish exactly for $j \equiv \pm 1$; and $d \neq 0$ for $k \neq 4$. For $k = 4$ use Proposition 3.4 with $a = b = 2$; -1 is a primitive square root of unity when the characteristic is not 2. □

Remark 3.7. The fields of Proposition 3.6 include every finite field $\mathbb{F}_{p^s}$ with $p \nmid k$ and $k \mid p^s - 1$, since the multiplicative group of $\mathbb{F}_{p^s}$ is cyclic of order $p^s - 1$. So for each odd k the cycle C_k is tight over a finite field of characteristic 2, which is the case used for Kneser graphs below. In characteristic $p \mid k$, the polynomial $x^k - 1$ has repeated roots, the circulant is no longer diagonalisable, and the argument does not apply.

3.5. **Kneser graphs.** The Kneser graph $K(n, 2)$ has the two-element subsets of $[n] = \{1, \ldots, n\}$ as vertices, two of them adjacent when they are disjoint. For $n = 5$ it is the Petersen graph.

Proposition 3.8. (i) *For $n \geq 6$, $\gamma_{\mathrm{gr}}(K(n,2)) = n - 1$, and $\gamma_{\mathrm{gr}}(K(5,2)) = 5$.*

(ii) *If $n \geq 6$ is even, $K(n,2)$ is $\mathbb{F}_2$-tight.*

(iii) *If $n \geq 7$ is odd, $K(n,2)$ is $\mathbb{F}_p$-tight for every odd prime p dividing $n - 4$; such a prime exists.*

(iv) *The Petersen graph $K(5,2)$ is $\mathbb{F}$-tight over every field of characteristic different from 2.*

Proof. Lower bounds. The $n - 1$ sets $\{1, j\}$, $2 \leq j \leq n$, pairwise intersect, so they form an independent set, and listing them gives a dominating sequence in which each set footprints itself. For $n = 5$ the sequence

$$\{1,2\},\ \{1,3\},\ \{2,4\},\ \{1,5\},\ \{1,4\}$$

has the footprints $\{1,2\}$, $\{1,3\}$, $\{1,5\}$, $\{2,3\}$, $\{1,4\}$, as one checks from the rule that a set dominates itself and the three sets disjoint from it.

The matrix identity. Let U be the $\binom{n}{2} \times n$ incidence matrix, $U_{S,x} = 1$ if $x \in S$. Over the integers,

$$(UU^{\mathsf{T}})_{ST} = |S \cap T| = \begin{cases} 2, & S = T, \\ 1, & |S \cap T| = 1, \\ 0, & S \cap T = \emptyset. \end{cases}$$

Writing A_0 for the adjacency matrix of $K(n,2)$ and A_1 for the matrix of the relation $|S \cap T| = 1$, we have $UU^{\mathsf{T}} = 2I + A_1$ and $J = I + A_0 + A_1$, hence

$$A_0 = J + I - UU^{\mathsf{T}}. \tag{3.1}$$

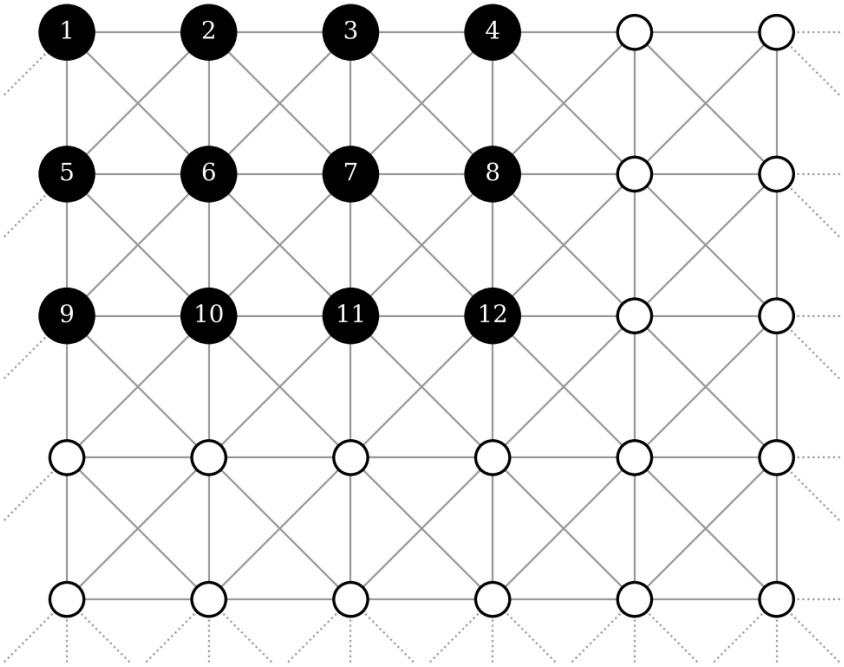


FIGURE 3. The strong torus $C_5 \boxtimes C_6$, drawn as a 5×6 grid whose opposite sides are identified (dotted edges wrap around). The $12 = (5-2)(6-2)$ marked vertices form a longest dominating sequence; by Theorem B no dominating sequence is longer.

(iii) and (iv). Let p be an odd prime, or let $\mathbb{F}$ have characteristic different from 2. Every row of U has two ones, so $U\mathbf{1}_n = 2 \cdot \mathbf{1}$, and $J = \mathbf{1}\mathbf{1}^{\mathsf{T}} = \frac{1}{4}U\mathbf{1}_n\mathbf{1}_n^{\mathsf{T}}U^{\mathsf{T}} = \frac{1}{4}UJ_nU^{\mathsf{T}}$. With (3.1),

$$A_0 - I = J - UU^{\mathsf{T}} = U(\tfrac{1}{4}J_n - I_n)U^{\mathsf{T}}.$$

The matrix $A_0 - I$ has diagonal entries $-1 \neq 0$, entries 1 on disjoint pairs and 0 on intersecting pairs, so $A_0 - I \in \mathcal{S}_{\mathbb{F}}[K(n,2)]$, and $\operatorname{rank}(A_0 - I) \leq \operatorname{rank}(\frac{1}{4}J_n - I_n) \leq n$. For $n = 5$ this gives $\operatorname{mr}_{\mathbb{F}}(K(5,2)) \leq 5 = \gamma_{\mathrm{gr}}(K(5,2))$, which is (iv). For (iii), let $n \geq 7$ be odd and p an odd prime dividing $n-4$; one exists because $n-4 \geq 3$ is odd. Then $n \equiv 4 \not\equiv 0 \pmod{p}$, so $\mathbb{F}_p^n$ is the direct sum of the line spanned by $\mathbf{1}_n$ and the hyperplane of vectors with coordinate sum 0. The matrix $\frac{1}{4}J_n - I_n$ acts on the line by $\frac{n}{4} - 1 = 0$ and on the hyperplane by -1. So its rank is $n-1$, and $\operatorname{mr}_{\mathbb{F}_p}(K(n,2)) \leq n - 1 = \gamma_{\mathrm{gr}}(K(n,2))$.

(ii). Let $n \geq 6$ be even and work over $\mathbb{F}_2$. Reducing (3.1) modulo 2 gives $A_0 + I = J + UU^{\mathsf{T}} = WW^{\mathsf{T}}$, where $W = (U \mid \mathbf{1})$ is the $\binom{n}{2} \times (n+1)$ matrix obtained by appending a column of ones. The matrix $A_0 + I$ lies in $\mathcal{S}_{\mathbb{F}_2}[K(n,2)]$. For any matrix W,

$$\operatorname{rank}(WW^{\mathsf{T}}) = \dim W(\operatorname{im} W^{\mathsf{T}}) = \operatorname{rank} W - \dim\bigl(\operatorname{im} W^{\mathsf{T}} \cap \ker W\bigr).$$

The vector $z = (\mathbf{1}_n, 0) \in \mathbb{F}_2^{n+1}$ lies in $\ker W$, since $Wz = U\mathbf{1}_n = 2 \cdot \mathbf{1} = 0$; this also shows $\operatorname{rank} W \leq n$. We claim that $z \in \operatorname{im} W^{\mathsf{T}}$, the row space of W. The row of W indexed by

$S = \{x, y\}$ is $(e_x + e_y, 1)$, so the sum of the rows over the edge set E of a graph Γ on $[n]$ is $(\sum_x \deg_\Gamma(x) e_x,\ |E|)$ modulo 2. It suffices to find a graph Γ on $[n]$ in which every vertex has odd degree and the number of edges is even. If $n \equiv 0 \pmod 4$, a perfect matching has $n/2$ edges, an even number. If $n \equiv 2 \pmod 4$, take a star with centre 1 and leaves $2, 3, 4$ together with a perfect matching of $\{5, \dots, n\}$; this has $3 + (n-4)/2$ edges, which is even because $(n-4)/2$ is odd. Hence $\dim(\operatorname{im} W^{\mathsf{T}} \cap \ker W) \geq 1$ and $\operatorname{rank}(A_0 + I) \leq n - 1 = \gamma_{\mathrm{gr}}(K(n,2))$.

Upper bound in (i). For $n \geq 6$, (ii) and (iii) give $\gamma_{\mathrm{gr}}(K(n,2)) \leq \mathrm{mr}_{\mathbb{F}}(K(n,2)) \leq n - 1$ for a suitable field. For $n = 5$, (iv) gives $\gamma_{\mathrm{gr}} \leq 5$. □

The equality $\gamma_{\mathrm{gr}}(K(n,2)) = n - 1$ for $n \geq 6$ was proved by Brešar, Kos and Torres [12] by a combinatorial argument; the matrices above give a two-line upper bound once the identity (3.1) is written down. The value 5 for the Petersen graph agrees with Lin's computation [13, Example 3.2] over the reals.

Remark 3.9. For even n the field $\mathbb{F}_2$ cannot be replaced by a field of characteristic 0: over $\mathbb{Q}$ the matrix $A_0 - I$ has rank n, since $\frac{1}{4}J_n - I_n$ is invertible when $n \neq 4$, and every matrix $aA_0 + bI$ in the Bose–Mesner algebra of the Johnson scheme has rank at least n for $n \geq 6$. We do not know whether $\mathrm{mr}_{\mathbb{C}}(K(n,2)) = n - 1$ for some $n \geq 6$.

4. Exact values for strong products

Proof of Theorem B. By Propositions 3.3, 3.4 and 3.6, every path, cycle and complete bipartite graph is $\mathbb{C}$-tight, and so is every complete graph. Corollary 2.5 gives the product formula for any strong product of such graphs, and in particular

$$\gamma_{\mathrm{gr}}\Big(\boxtimes_{i=1}^{r} C_{k_i}\Big) = \prod_{i=1}^{r} \gamma_{\mathrm{gr}}(C_{k_i}) = \prod_{i=1}^{r} (k_i - 2).$$

For Kneser graphs, let $n \geq 6$ be even and $k \geq 3$ odd. Choose s with $k \mid 2^s - 1$; then $\mathbb{F}_{2^s}$ contains $\mathbb{F}_2$ and a primitive k-th root of unity, so $K(n,2)$ (Proposition 3.8(ii) and (2.1)) and C_k (Proposition 3.6) are both $\mathbb{F}_{2^s}$-tight, and Corollary 2.5 gives $\gamma_{\mathrm{gr}}(K(n,2) \boxtimes C_k) = (n-1)(k-2)$. If $n, m \geq 6$ are even, both Kneser graphs are $\mathbb{F}_2$-tight and $\gamma_{\mathrm{gr}}(K(n,2) \boxtimes K(m,2)) = (n-1)(m-1)$. □

The proof gives more. We collect the cases in one statement.

Corollary 4.1. *Let $G_1, \dots, G_r$ be graphs each of which is a complete graph, an edgeless graph, a path, a cycle, a complete bipartite graph or a cocktail party graph. Then $\gamma_{\mathrm{gr}}(G_1 \boxtimes \cdots \boxtimes G_r) = \prod_i \gamma_{\mathrm{gr}}(G_i)$, where*

$$\gamma_{\mathrm{gr}}(K_n) = 1, \qquad \gamma_{\mathrm{gr}}(\overline{K_n}) = n, \qquad \gamma_{\mathrm{gr}}(P_m) = m - 1,$$
$$\gamma_{\mathrm{gr}}(C_k) = k - 2, \qquad \gamma_{\mathrm{gr}}(K_{a,b}) = \max(a, b), \qquad \gamma_{\mathrm{gr}}(CP(s)) = 2.$$

The same holds if some of the factors are Petersen graphs. It also holds for factors $K(n,2)$ with n even together with cycles of odd length and complete graphs, and for factors $K(n,2)$ with n odd together with cycles C_k such that some odd prime p divides $n-4$ for every such n and does not divide any such k.

Proof. All the graphs in the first list are $\mathbb{C}$-tight, by the propositions of Section 3; the Petersen graph is $\mathbb{C}$-tight by Proposition 3.8(iv). For the Kneser factors with n even use a finite field of characteristic 2 containing primitive roots of unity of all the odd cycle lengths, and for n odd a finite field of characteristic p containing primitive roots of unity of all the cycle lengths, which exists because p divides none of them. □

TABLE 1. Grundy domination numbers of strong products given by Corollary 4.1. The column "field" names a field over which all factors are tight.

product	γ_{gr}	field
$P_k \boxtimes P_\ell$	$(k-1)(\ell-1)$	$\mathbb{R}$
$P_k \boxtimes C_\ell$	$(k-1)(\ell-2)$	$\mathbb{R}$
$C_k \boxtimes C_\ell$	$(k-2)(\ell-2)$	$\mathbb{C}$
$C_{k_1} \boxtimes \cdots \boxtimes C_{k_r}$	$\prod(k_i-2)$	$\mathbb{C}$
$C_k \boxtimes K_{a,b}$, $a \geq b$	$a(k-2)$	$\mathbb{C}$
$C_k \boxtimes CP(s)$	$2(k-2)$	$\mathbb{C}$
Petersen $\boxtimes C_k$	$5(k-2)$	$\mathbb{C}$
Petersen $\boxtimes$ Petersen	25	$\mathbb{C}$
$K(n,2) \boxtimes C_k$, n even, k odd	$(n-1)(k-2)$	$\mathbb{F}_{2^s}$
$K(n,2) \boxtimes K(m,2)$, n,m even	$(n-1)(m-1)$	$\mathbb{F}_2$
$K(7,2) \boxtimes K(10,2)$	54	$\mathbb{F}_3$
$K(9,2) \boxtimes C_k$, $5 \nmid k$	$8(k-2)$	$\mathbb{F}_{5^s}$

Table 1 lists some of the values. The torus entries settle the case left open in [7, Corollary 29], where the bound $(k-2)(\ell-1)$ was proved; the cylinder and grid entries recover [7, Corollaries 24 and 28].

For $K(7,2)$ and $K(10,2)$ the common odd prime is 3, which divides $7-4$ and $10-4$. That the even graph $K(10,2)$ is also tight over $\mathbb{F}_3$ follows from the next proposition, whose proof is the argument for part (iii) of Proposition 3.8.

Proposition 4.2. *Let $n \geq 6$ and let p be an odd prime with $n \equiv 4 \pmod p$. Then $K(n,2)$ is $\mathbb{F}_p$-tight.*

Proof. The proof of Proposition 3.8(iii) used only that p is odd and $n \equiv 4 \not\equiv 0 \pmod p$. □

Remark 4.3. For $3 \leq k \leq \ell \leq 7$ the value $\gamma_{\mathrm{gr}}(C_k \boxtimes C_\ell) = (k-2)(\ell-2)$ was also confirmed by an exhaustive computation of the longest dominating sequence (Section 8); for instance $\gamma_{\mathrm{gr}}(C_7 \boxtimes C_7) = 25$ on 49 vertices, and $\gamma_{\mathrm{gr}}(\text{Petersen} \boxtimes C_5) = 15$. The proofs do not use these computations.

4.1. **Strong products with a fixed tight factor.** Corollary 2.5 needs both factors tight. When only one factor is tight, the rank bound still gives an inequality.

Proposition 4.4. *If G is $\mathbb{F}$-tight, then for every graph H,*

$$\gamma_{\mathrm{gr}}(G)\,\gamma_{\mathrm{gr}}(H) \leq \gamma_{\mathrm{gr}}(G \boxtimes H) \leq \gamma_{\mathrm{gr}}(G)\ \mathrm{mr}_{\mathbb{F}}(H).$$

Proof. The lower bound is Lemma 2.3, and the upper bound is Theorem A with $\mathrm{mr}_{\mathbb{F}}(G) = \gamma_{\mathrm{gr}}(G)$. □

So for a tight G the gap in Conjecture 1.1 is controlled by the gap $\mathrm{mr}_{\mathbb{F}}(H) - \gamma_{\mathrm{gr}}(H)$ of the other factor; the octahedron is tight, and the graphs H_N of Theorem C have a large gap.

5. The strong product conjecture fails

By Corollary 2.5, a counterexample to Conjecture 1.1 must contain, for each characteristic, a factor that is not tight. The first construction uses two factors bounded over different fields, so that no common field is available; the second uses a tight factor, the octahedron, and a factor that is tight over no field at all.

5.1. **Unbounded ratios.**

Theorem 5.1. *For $m \geq 5$ let A_m and B_m be the graphs on the five-element subsets of $[m]$ in which distinct sets S, T are adjacent when $|S \cap T|$ is odd, respectively when $|S \cap T| = 2$. Then*

$$\gamma_{\mathrm{gr}}(A_m) \leq m, \qquad \gamma_{\mathrm{gr}}(B_m) \leq \binom{m}{2}, \qquad \gamma_{\mathrm{gr}}(A_m \boxtimes B_m) \geq \binom{m}{5}.$$

Consequently

$$\frac{\gamma_{\mathrm{gr}}(A_m \boxtimes B_m)}{\gamma_{\mathrm{gr}}(A_m)\,\gamma_{\mathrm{gr}}(B_m)} \;\geq\; \frac{(m-2)(m-3)(m-4)}{60\,m} \;\longrightarrow\; \infty.$$

Proof. Let U be the incidence matrix between the five-element subsets of $[m]$ and the elements of $[m]$. Over $\mathbb{F}_2$ the matrix UU^{T} has entries $|S \cap T| \bmod 2$. The diagonal entries are $5 \equiv 1$, and an off-diagonal entry is nonzero exactly when $|S \cap T|$ is odd. So $UU^{\mathsf{T}} \in \mathcal{S}_{\mathbb{F}_2}[A_m]$, its rank is at most m, and Lemma 2.1 gives $\gamma_{\mathrm{gr}}(A_m) \leq m$.

Let V be the incidence matrix between five-element subsets and two-element subsets of $[m]$, with $V_{S,P} = 1$ when $P \subseteq S$. Over $\mathbb{F}_3$, $(VV^{\mathsf{T}})_{ST} = \binom{|S\cap T|}{2} \bmod 3$, the number of pairs inside $S \cap T$. For $t = 0, 1, \ldots, 5$ the binomial coefficients $\binom{t}{2} = 0, 0, 1, 3, 6, 10$ are congruent to $0, 0, 1, 0, 0, 1$ modulo 3, so the nonzero entries are those with $S = T$ or $|S \cap T| = 2$. Hence $VV^{\mathsf{T}} \in \mathcal{S}_{\mathbb{F}_3}[B_m]$ has rank at most $\binom{m}{2}$, and $\gamma_{\mathrm{gr}}(B_m) \leq \binom{m}{2}$.

Finally, the diagonal $\{(S, S)\}$ is an independent set of $A_m \boxtimes B_m$: for $S \neq T$ adjacency of (S, S) and (T, T) would require $|S \cap T|$ to be odd and equal to 2. Listing an independent set in any order gives a dominating sequence, each vertex footprinting itself. This gives the lower bound, and the ratio is

$$\binom{m}{5} \Big/ \Big(m\binom{m}{2}\Big) = \frac{(m-2)(m-3)(m-4)}{60m}. \qquad \square$$

For $m = 13$ the product has a dominating sequence of length 1287, while $\gamma_{\mathrm{gr}}(A_{13})\gamma_{\mathrm{gr}}(B_{13}) \leq 13 \cdot 78 = 1014$. The two factor bounds are over $\mathbb{F}_2$ and $\mathbb{F}_3$. Over a common field the product of the two matrices would bound the product graph, by Theorem A, so the change of field is forced. The construction is in the spirit of the intersection-parity arguments of Alon for the Shannon capacity of a disjoint union [1], with the difference that Lemma 2.1 needs the nonzero entries to be exactly the closed neighbourhoods, not merely the diagonal.[6]

5.2. **The octahedron.** The octahedron is the cocktail party graph $K_{2,2,2} = CP(3)$. Write its vertices as $g_0, g_1, g_2, x_0, x_1, x_2$, with x_t the antipode of g_t, and read indices modulo 3. By Proposition 3.5, $\gamma_{\mathrm{gr}}(K_{2,2,2}) = 2$.

Proposition 5.2. *Let H be a graph with vertices $v_0, \ldots, v_{N-1}$ and let $t : V(H) \to \mathbb{Z}/3$ satisfy $t(v_i) = t(v_j) + 1$ for every edge v_iv_j with $i < j$. Then*

$$s_j = (g_{t(v_j)}, v_j), \qquad j = 0, 1, \ldots, N-1,$$

is a dominating sequence of $K_{2,2,2} \boxtimes H$, the vertex s_j footprinting $f_j = (x_{t(v_j)+1}, v_j)$. Hence $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H) \geq N$.

Proof. The vertex $x_{t(v_j)+1}$ is not the antipode $x_{t(v_j)}$ of $g_{t(v_j)}$, so it lies in $N[g_{t(v_j)}]$, and $f_j \in N[s_j]$. Let $i < j$ and suppose $f_j \in N[s_i]$. Then $x_{t(v_j)+1} \in N[g_{t(v_i)}]$, so $t(v_i) \neq t(v_j) + 1$; and $v_j \in N[v_i]$ with $v_i \neq v_j$, so v_iv_j is an edge and $t(v_i) = t(v_j) + 1$. This is a contradiction. $\square$

[6]For an independent set the diagonal alone would suffice. The difference between bounding the independence number and bounding the Grundy domination number is that the off-diagonal support must match the edges.

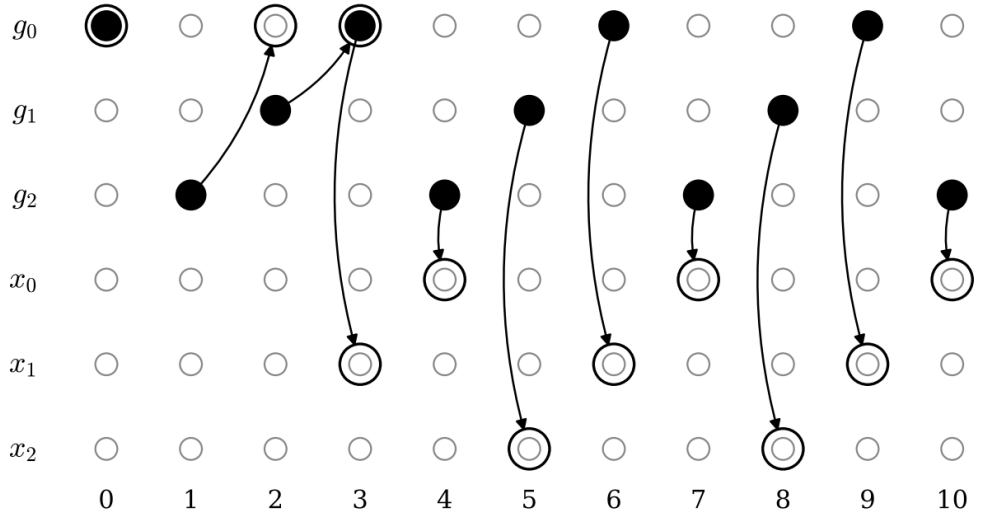


FIGURE 4. The strong product $K_{2,2,2} \boxtimes H_{11}$ drawn as six layers, one for each vertex of the octahedron, over the 11 vertices of H_{11}. The dominating sequence of Proposition 5.2 uses one vertex of each column, in the layers $g_0, g_2, g_1, g_0, \ldots$, and each circled vertex is the footprint of the term joined to it by an arrow (the first term is its own footprint).

The colouring rotates the three layers x_0, x_1, x_2 of the octahedron along the sequence, so that every new vertex reaches a footprint in a layer that the earlier vertices of its column have left uncovered (Figure 4). For H_N, the graph on $\{0, \ldots, N-1\}$ with $u \sim v$ when $|u - v| \equiv 1 \pmod 3$, the colouring $t(v) = -v \bmod 3$ has the required property: if $u < v$ are adjacent, then $v - u \equiv 1$, so $t(u) \equiv t(v) + 1$. Therefore

$$\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \geq N \qquad (N \geq 1). \tag{5.1}$$

5.3. **The Grundy domination number of H_N.** Write $a_i = 3i$, $b_i = 3i + 1$ and $c_i = 3i + 2$ for the vertices of H_N. Two vertices of one residue class are never adjacent, and comparing differences modulo 3 gives

$$a_i \sim b_j \iff j \geq i, \qquad b_i \sim c_j \iff j \geq i, \qquad c_i \sim a_j \iff j > i. \tag{5.2}$$

For example $b_j - a_i = 3(j - i) + 1$, which is congruent to 1 when $j \geq i$ and has absolute value $3(i - j) - 1 \equiv 2$ when $j < i$; the other two cases are the same computation.

Lemma 5.3. *For $N \geq 4$, $\gamma_{\mathrm{gr}}(H_N) \geq \lfloor N/3 \rfloor + 2$.*

Proof. Let $q = \lfloor N/3 \rfloor$. By Lemma 3.1 it suffices to find a dominating sequence of length $q + 2$ in an induced subgraph $H_{N'}$ with $N' \leq N$. For $q = 1$, H_4 is the path $0, 1, 2, 3$ and $(0, 1, 2)$ has the footprints $0, 2, 3$. For $q = 2$, the sequence $(0, 5, 1, 2)$ of H_6 has the footprints $0, 5, 2, 3$. For $q \geq 3$ consider in H_{3q} the sequence

$$a_0,\ a_1,\ c_0,\ c_1,\ \ldots,\ c_{q-2},\ b_0,$$

of length $q + 2$. By (5.2): a_0 footprints itself; a_1 footprints itself, since it is not adjacent to a_0; c_0 is adjacent to a_2, which neither a_0 nor a_1 dominates, so a_2 is a footprint; for $1 \leq m \leq q - 2$ the vertex c_m is adjacent to no earlier vertex (the only earlier vertex adjacent to a c is a_1,

adjacent to c_0 alone), so it footprints itself; and b_0 is adjacent to c_{q-1}, which no earlier vertex dominates. □

The upper bound comes from the structure of the vertices not dominated by the first vertex of a sequence.

Lemma 5.4. *Let $q \geq 1$ and let z be a vertex of a graph Γ whose non-neighbours are $2q$ vertices $X_0, \ldots, X_{q-1}$, $Y_0, \ldots, Y_{q-1}$, with no edges among the X_i, none among the Y_j, and $X_i \sim Y_j$ exactly when $i \geq j$. Suppose that every neighbour w of z has an index $k \in \{0, \ldots, q\}$ with*

$$N[w] \cap \{X_i, Y_j\} = \{X_i : i < k\} \cup \{Y_j : j \geq k\}.$$

Then every dominating sequence of Γ beginning with z has length at most $q + 2$.

Proof. Let $(z, v_1, \ldots, v_m)$ be a dominating sequence. After z, the undominated vertices are the X_i and Y_j, which induce a half graph containing the perfect matching $\{X_iY_i\}$. We use the following counting principle: if after some step exactly u vertices are undominated, at most u further steps follow, since each step dominates a new vertex.

Let τ be the first index for which v_τ is a neighbour of z, or v_τ is an X or a Y adjacent to an earlier v_i. If there is no such index, the v_i form an independent set of the half graph, which contains at most one end of each matching edge, so $m \leq q$. Otherwise let I and J be the sets of indices of the X's and Y's among $v_1, \ldots, v_{\tau-1}$, with $r = |I|$ and $s = |J|$, so $\tau - 1 = r + s$. Since no two of them are adjacent, $i < j$ for all $i \in I$ and $j \in J$.

Case 1: $v_\tau = X_i$ is adjacent to an earlier Y_j. Then $J \neq \emptyset$ and $b = \min J \leq i$, and all indices in I are less than b. After step τ an X-vertex is dominated if it was selected or is adjacent to a selected Y, that is, if its index lies in $I \cup \{i\}$ or is at least b; the undominated X-vertices are the $b - r$ vertices with index in $\{0, \ldots, b-1\} \setminus I$. A Y-vertex is dominated if it was selected or is adjacent to a selected X, and the largest selected X-index is i; with $s_0 = |J \cap \{b, \ldots, i\}|$, the undominated Y-vertices number $(q - 1 - i) - (s - s_0)$. By the counting principle,

$$m \leq \tau + (b - r) + (q - 1 - i) - (s - s_0) = q + b - i + s_0 \leq q + 1,$$

because $s_0 \leq i - b + 1$.

Case 2: $v_\tau = Y_j$ is adjacent to an earlier X_i. The map $X_i \mapsto Y_{q-1-i}$, $Y_j \mapsto X_{q-1-j}$ preserves the half graph, since $q - 1 - j \geq q - 1 - i$ exactly when $i \geq j$, and before step τ only X's and Y's were selected; the count of Case 1 applies verbatim and gives $m \leq q + 1$.

Case 3: $v_\tau = w$ is a neighbour of z with index k. After step τ every undominated vertex lies in $W_k = \{X_i : i \geq k\} \cup \{Y_j : j < k\}$, a set of q vertices. We claim that at least $r + s$ vertices of W_k were dominated before step τ. Send a selected X_i to X_i if $i \geq k$ and to Y_i if $i < k$, and a selected Y_j to Y_j if $j < k$ and to X_j if $j \geq k$. Each image lies in W_k and is dominated by the selected vertex, which is either the image itself or its partner in the matching. Distinct selections on one side have distinct images, and a selection X_i and a selection Y_j could share an image only if $i = j$, which is impossible because $i < j$. So the map is injective, at most $q - r - s$ vertices remain undominated after step τ, and

$$m \leq \tau + q - r - s = q + 1.$$

In all cases the sequence $(z, v_1, \ldots, v_m)$ has length at most $q + 2$. □

Lemma 5.5. *For every $q \geq 1$, $\gamma_{\mathrm{gr}}(H_{3q+2}) \leq q + 2$.*

Proof. Let $n = 3q + 2$. For $1 \leq d < n$ we have $d \equiv 1 \pmod 3$ if and only if $n - d \equiv 1 \pmod 3$, because $n \equiv 2$. Hence $u \sim v$ in H_n exactly when $(u - v) \bmod n \equiv 1 \pmod 3$, and translation modulo n is an automorphism. A dominating sequence remains one under an automorphism, so we may assume that it begins with $z = 0$. The neighbours of 0 are $b_0, \ldots, b_q$. Its non-neighbours

are $X_i = a_{i+1}$ $(0 \le i < q)$ and $Y_j = c_j$ $(0 \le j < q)$. By (5.2), $X_i = a_{i+1} \sim c_j$ exactly when $i+1 > j$, that is, $i \ge j$; and b_k is adjacent to a_{i+1} exactly when $i < k$ and to c_j exactly when $j \ge k$. Lemma 5.4 applies. □

Theorem 5.6. *For $N \ge 4$, $\gamma_{\mathrm{gr}}(H_N) = \lfloor N/3 \rfloor + 2$ and $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \ge N$. Conjecture 1.1 fails for $K_{2,2,2}$ and H_N exactly when $N > 2\gamma_{\mathrm{gr}}(H_N)$, that is, for $N \in \{11, 13, 14\}$ and all $N \ge 15$.*

Proof. With $q = \lfloor N/3 \rfloor$, H_N is an induced subgraph of H_{3q+2}, so Lemmas 5.3, 3.1 and 5.5 give $\gamma_{\mathrm{gr}}(H_N) = q + 2$. The lower bound for the product is (5.1). Write $N = 3q + e$ with $e \in \{0, 1, 2\}$; then $N - 2(q+2) = q + e - 4 > 0$ exactly when $q \ge 5 - e$, which gives the stated values.[7] □

The smallest of these counterexamples is $K_{2,2,2} \boxtimes H_{11}$, with 66 vertices: $\gamma_{\mathrm{gr}}(K_{2,2,2}) = 2$ and $\gamma_{\mathrm{gr}}(H_{11}) = 5$, while the product has the dominating sequence $(g_{-j \bmod 3}, j)$, $0 \le j \le 10$, of length 11.

Corollary 5.7. *For $N \in \{11, 13, 14\}$ and $N \ge 15$, the graph H_N is not tight over any field: $\gamma_{\mathrm{gr}}(H_N) < \mathrm{mr}_{\mathbb{F}}(H_N)$ for every field $\mathbb{F}$.*

Proof. Suppose H_N were $\mathbb{F}$-tight. Let $\mathbb{F}' \supseteq \mathbb{F}$ be a finite extension with at least five elements. By (2.1), H_N is $\mathbb{F}'$-tight, and $K_{2,2,2}$ is $\mathbb{F}'$-tight by Proposition 3.5. Corollary 2.5 would give $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) = 2\gamma_{\mathrm{gr}}(H_N) < N$, contradicting (5.1). □

So the octahedron, although it is the factor with the small Grundy number, is not the source of the failure; the graphs H_N are. In the language of Proposition 4.4, for every field $\mathbb{F}$ and every N as above,

$$\mathrm{mr}_{\mathbb{F}}(H_N) \;\ge\; \tfrac{1}{2}\,\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N) \;\ge\; \tfrac{N}{2},$$

which is roughly $\frac{3}{2}\gamma_{\mathrm{gr}}(H_N)$. With the octahedron fixed, the ratio $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N)/(2\gamma_{\mathrm{gr}}(H_N))$ tends to $3/2$; unbounded ratios require both factors to vary, as in Theorem 5.1.

6. Direct products and Grundy total domination

Throughout this section all graphs are without isolated vertices, so that open neighbourhoods are nonempty and γ^t_{gr} is defined. A vertex u is a *total footprint* of v_i in a sequence $(v_1, \ldots, v_k)$ if $u \in N(v_i) \setminus \bigcup_{j<i} N(v_j)$, and the sequence is a total dominating sequence if every term has a total footprint.

6.1. **Totally tight graphs.**

Lemma 6.1. *If H is an induced subgraph of G and neither graph has isolated vertices, then $\gamma^t_{\mathrm{gr}}(H) \le \gamma^t_{\mathrm{gr}}(G)$. The Grundy total domination number of a disjoint union is the sum of those of the components.*

Proof. Let $(v_1, \ldots, v_k)$ be a total dominating sequence of H with total footprints $u_1, \ldots, u_k$. For vertices u, v of H, $u \in N_G(v)$ if and only if $u \in N_H(v)$, since H is induced. Hence $u_i \in N_G(v_i)$ and $u_i \notin N_G(v_j)$ for $j < i$, and the same sequence is a total dominating sequence of G. For a disjoint union, the subsequence of a total dominating sequence lying in one component is a total dominating sequence of that component, and conversely the concatenation of sequences of the components is one of the union. □

Recall that the matching number $\nu(F)$ of a graph F is the largest number of pairwise disjoint edges.

[7] "Exactly" refers to the inequality $N > 2\gamma_{\mathrm{gr}}(H_N)$; for $N = 12$ the lower bound (5.1) equals $2\gamma_{\mathrm{gr}}(H_{12})$ and gives no counterexample, and the exact value $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_{12}) = 12$ found by computation (Section 8) shows that there is none.

Proposition 6.2. *Let F be a forest without isolated vertices. Then $\gamma^t_{\mathrm{gr}}(F) = 2\nu(F)$, and F is totally $\mathbb{F}$-tight over every field $\mathbb{F}$: its adjacency matrix has rank $2\nu(F)$ over $\mathbb{F}$.*

Proof. We prove $\gamma^t_{\mathrm{gr}}(F) \geq 2\nu(F)$ by induction on $|V(F)|$. By Lemma 6.1 it suffices to treat a tree T. If $T = K_2$ the claim is $\gamma^t_{\mathrm{gr}}(K_2) = 2$, which is clear. Otherwise let $\ell s \cdots$ be a longest path of T, beginning at the leaf ℓ. Every neighbour of s other than its successor p on the path is a leaf, for otherwise the path could be lengthened. Let L be the set of leaves adjacent to s and $T' = T - (\{s\} \cup L)$, with the vertex p removed as well if it becomes isolated. Every matching of T contains at most one edge meeting $\{s\} \cup L$, so $\nu(T) \leq \nu(T') + 1$. Let $(x_1, \ldots, x_m)$ be a total dominating sequence of T' of length $\gamma^t_{\mathrm{gr}}(T')$ (empty if T' is empty). We claim that

$$(\ell,\ x_1, \ldots, x_m,\ s)$$

is a total dominating sequence of T. The first term has total footprint s. A total footprint $y_i \in V(T')$ of x_i in T' is not adjacent to ℓ, because $N(\ell) = \{s\}$, and is not adjacent in T to any earlier x_j, because T' is induced. The last term has total footprint ℓ, since s is the only neighbour of ℓ. By induction $\gamma^t_{\mathrm{gr}}(T) \geq 2 + 2\nu(T') \geq 2\nu(T)$.

Conversely, let A be the adjacency matrix of F, a symmetric matrix with zero diagonal, so $A \in \mathcal{S}_{\mathbb{F}}(F)$ for every field $\mathbb{F}$. The rank of a symmetric matrix over any field equals the order of its largest nonsingular principal submatrix. A principal submatrix of A is the adjacency matrix of an induced subforest $F[S]$, and its determinant is $\pm$ the number of perfect matchings of $F[S]$, because a permutation contributes to the determinant only if all of its cycles have length two when the graph has no cycles. A forest has at most one perfect matching, so $\det A[S] = \pm 1$ if $F[S]$ has a perfect matching and 0 otherwise, in every characteristic. The largest such S is the vertex set of a maximum matching, so $\operatorname{rank}_{\mathbb{F}} A = 2\nu(F)$. With Lemma 2.1, $2\nu(F) \leq \gamma^t_{\mathrm{gr}}(F) \leq \mathrm{mr}^{\circ}_{\mathbb{F}}(F) \leq 2\nu(F)$. □

By König's theorem $\nu(F)$ is also the least size $\beta(F)$ of a vertex cover of F, and the value $\gamma^t_{\mathrm{gr}}(T) = 2\beta(T)$ for trees is due to Brešar, Kos, Nasini and Torres [11]; the proof above is included because the rank statement is what Theorem 6.4 needs. For the path, $\gamma^t_{\mathrm{gr}}(P_m) = 2\lfloor m/2 \rfloor$.

Proposition 6.3. (i) *For $a, b \geq 1$, $\gamma^t_{\mathrm{gr}}(K_{a,b}) = \mathrm{mr}^{\circ}_{\mathbb{F}}(K_{a,b}) = 2$ over every field.*

(ii) *For $n \geq 2$, $\gamma^t_{\mathrm{gr}}(K_n) = 2$, and K_n is totally $\mathbb{F}$-tight over every field with at least n elements.*

(iii) *For $n \geq 3$, $\gamma^t_{\mathrm{gr}}(C_n) = 2\lfloor (n-1)/2 \rfloor$, that is, $n-1$ for odd n and $n-2$ for even n. The cycle C_n is totally $\mathbb{R}$-tight for even n and totally $\mathbb{C}$-tight for every n.*

Proof. (i) The matrix $\left(\begin{smallmatrix} 0 & J \\ J & 0 \end{smallmatrix}\right)$ lies in $\mathcal{S}_{\mathbb{F}}(K_{a,b})$ and has rank 2. A vertex of one side and then a vertex of the other side form a total dominating sequence, and nothing can follow, since together they dominate every vertex.

(ii) Choose distinct $x_v \in \mathbb{F}$ and set $M_{uv} = x_u - x_v$. The diagonal vanishes and the off-diagonal entries do not, and $M = x\mathbf{1}^{\mathsf{T}} - \mathbf{1}x^{\mathsf{T}}$ has rank at most 2. The sequence (v, w) with $v \neq w$ has total footprints w and v.

(iii) Deleting a vertex of C_n leaves an induced path P_{n-1}, so Lemma 6.1 and Proposition 6.2 give $\gamma^t_{\mathrm{gr}}(C_n) \geq 2\lfloor (n-1)/2 \rfloor$. For the upper bound let $\omega = e^{i\theta}$ and let A_ω be the Hermitian matrix with $(A_\omega)_{t,t+1} = 1$ and $(A_\omega)_{t+1,t} = 1$ for $0 \leq t \leq n-2$, $(A_\omega)_{n-1,0} = \omega$ and $(A_\omega)_{0,n-1} = \bar{\omega}$. It lies in $\mathcal{S}_{\mathbb{C}}(C_n)$. With $\alpha = e^{i\theta/n}$ and $D = \operatorname{diag}(\alpha^{-t})_{0 \leq t < n}$, one checks entry by entry that $DA_\omega D^{-1} = \alpha S + \bar{\alpha} S^{-1}$, where S is the permutation matrix with $S_{t,t+1} = 1$, indices modulo n; the eigenvalues of S are the n-th roots of unity, so the eigenvalues of A_ω are

$$2\cos\frac{\theta + 2\pi j}{n}, \qquad 0 \leq j \leq n-1.$$

TABLE 2. Values of γ^t_{gr} for direct products given by Theorem 6.4 and confirmed by the exact solver of Appendix B. Here T_7 is the complete binary tree with seven vertices, for which $\nu(T_7) = 2$.

product	vertices	γ^t_{gr}	product	vertices	γ^t_{gr}
$C_3 \times C_3$	9	4	$C_6 \times C_5$	30	16
$C_3 \times C_5$	15	8	$C_5 \times C_7$	35	24
$C_4 \times C_5$	20	8	$C_3 \times C_3 \times C_3$	27	8
$C_5 \times C_5$	25	16	$K_4 \times C_5$	20	8
$C_3 \times C_7$	21	12	$T_7 \times C_5$	35	16

If $n \equiv 0 \pmod 4$ take $\theta = 0$, and if $n \equiv 2 \pmod 4$ take $\theta = \pi$; in both cases exactly two of the angles $(\theta + 2\pi j)/n$ are odd multiples of $\pi/2$, and A_ω is a real matrix of rank $n-2$. If n is odd take $\theta = \pi/2$; then $(\pi/2 + 2\pi j)/n \equiv \pi/2 \pmod \pi$ means $4j + 1 \equiv n \pmod{2n}$ or $4j + 1 \equiv 3n \pmod{2n}$, and exactly one $j \in \{0, \dots, n-1\}$ satisfies one of these, so A_ω has rank $n-1$. □

Theorem 6.4. *Let $\mathbb{F}$ be a field. Direct products of totally $\mathbb{F}$-tight graphs are totally $\mathbb{F}$-tight, and for totally $\mathbb{F}$-tight graphs $G_1, \dots, G_r$,*

$$\gamma^t_{\mathrm{gr}}(G_1 \times \cdots \times G_r) = \prod_{i=1}^{r} \gamma^t_{\mathrm{gr}}(G_i).$$

In particular, for every $r \geq 1$, all $k_1, \dots, k_r \geq 3$ and all forests $F_1, \dots, F_s$ without isolated vertices,

$$\gamma^t_{\mathrm{gr}}(C_{k_1} \times \cdots \times C_{k_r} \times F_1 \times \cdots \times F_s) = \prod_{i=1}^{r} 2\Big\lfloor \frac{k_i - 1}{2} \Big\rfloor \prod_{j=1}^{s} 2\nu(F_j),$$

and factors K_n and $K_{a,b}$ may be added, each contributing a factor 2.

Proof. If $M \in \mathcal{S}_{\mathbb{F}}(G)$ and $M' \in \mathcal{S}_{\mathbb{F}}(H)$, then $M \otimes M'$ has its nonzero entries exactly at the pairs $((g,h),(g',h'))$ with $g' \in N(g)$ and $h' \in N(h)$, so $M \otimes M' \in \mathcal{S}_{\mathbb{F}}(G \times H)$ and $\mathrm{mr}^\circ_{\mathbb{F}}(G \times H) \leq \mathrm{mr}^\circ_{\mathbb{F}}(G)\,\mathrm{mr}^\circ_{\mathbb{F}}(H)$. Together with (1.3) and Lemma 2.1,

$$\gamma^t_{\mathrm{gr}}(G)\gamma^t_{\mathrm{gr}}(H) \leq \gamma^t_{\mathrm{gr}}(G \times H) \leq \mathrm{mr}^\circ_{\mathbb{F}}(G \times H) \leq \mathrm{mr}^\circ_{\mathbb{F}}(G)\,\mathrm{mr}^\circ_{\mathbb{F}}(H) = \gamma^t_{\mathrm{gr}}(G)\gamma^t_{\mathrm{gr}}(H)$$

for totally $\mathbb{F}$-tight G and H, and equality holds throughout. Induction on r gives the general case. By Propositions 6.2 and 6.3 all cycles, forests without isolated vertices, complete graphs and complete bipartite graphs are totally $\mathbb{C}$-tight, which gives the displayed formula. □

For two cycles Theorem 6.4 recovers the theorem of [6]; their theorem for a tree factor allows an arbitrary second factor and is not a consequence of ours. Table 2 lists values confirmed by exact computation.

6.2. **From closed to open neighbourhoods.** For a graph G let $\mathsf{B}(G)$ be the bipartite graph with vertex set $V(G) \times \{0,1\}$ in which (u,a) and (v,b) are adjacent when $a \neq b$ and $v \in N[u]$. Since $u \in N[u]$, the graph $\mathsf{B}(G)$ has no isolated vertices; it is the direct product $G^\circ \times K_2$, where G° is G with a loop at every vertex (Figure 5).

Proposition 6.5. *For all graphs G and H,*

(i) $\gamma^t_{\mathrm{gr}}(\mathsf{B}(G)) = 2\gamma_{\mathrm{gr}}(G)$;
(ii) $\mathsf{B}(G) \times \mathsf{B}(H)$ *is isomorphic to the disjoint union of two copies of* $\mathsf{B}(G \boxtimes H)$, *so that* $\gamma^t_{\mathrm{gr}}(\mathsf{B}(G) \times \mathsf{B}(H)) = 4\gamma_{\mathrm{gr}}(G \boxtimes H)$.

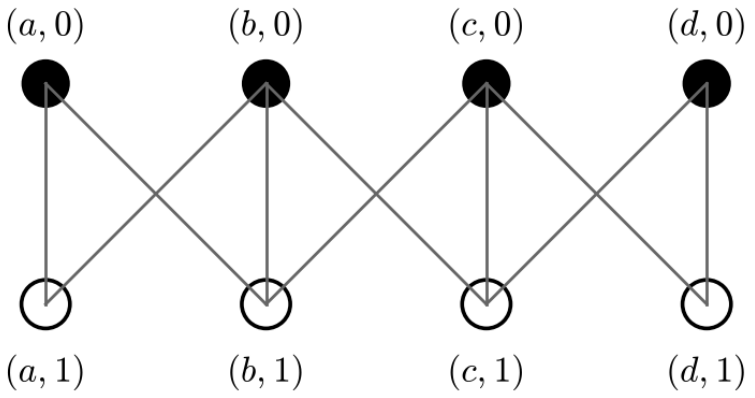


FIGURE 5. The bipartite double $\mathsf{B}(P_4)$ of the path $a\,b\,c\,d$: each vertex on the top row is joined to its closed neighbourhood on the bottom row. A dominating sequence of P_4, read forwards on the top row and its footprints read backwards on the bottom row, is a total dominating sequence of $\mathsf{B}(P_4)$ of twice the length.

Proof. (i) Let $(v_1, \dots, v_k)$ be a dominating sequence of G with footprints $u_1, \dots, u_k$. Then

$$((v_1, 0), \dots, (v_k, 0), (u_k, 1), (u_{k-1}, 1), \dots, (u_1, 1))$$

is a total dominating sequence of $\mathsf{B}(G)$. Indeed $(u_i, 1)$ is a total footprint of $(v_i, 0)$, because the neighbours of $(v_j, 0)$ are the vertices $(w, 1)$ with $w \in N[v_j]$, and $u_i \notin N[v_j]$ for $j < i$. The vertex $(v_i, 0)$ is a total footprint of $(u_i, 1)$, because $v_i \in N[u_i]$, the earlier terms on side 0 dominate only side 1, and the earlier terms on side 1 are the $(u_j, 1)$ with $j > i$, which are not adjacent to $(v_i, 0)$ because $u_j \notin N[v_i]$. So $\gamma^t_{\mathrm{gr}}(\mathsf{B}(G)) \geq 2\gamma_{\mathrm{gr}}(G)$.

Conversely let σ be a total dominating sequence of $\mathsf{B}(G)$. Terms on side 0 dominate only vertices of side 1 and vice versa. So the terms $(x_1, 0), \dots, (x_s, 0)$ of σ on side 0, in their order in σ, have total footprints $(y_1, 1), \dots, (y_s, 1)$ with $y_i \in N[x_i] \setminus \bigcup_{j<i} N[x_j]$; that is, $(x_1, \dots, x_s)$ is a dominating sequence of G and $s \leq \gamma_{\mathrm{gr}}(G)$. The same holds for side 1, so $|\sigma| \leq 2\gamma_{\mathrm{gr}}(G)$.

(ii) The vertices of $\mathsf{B}(G) \times \mathsf{B}(H)$ are the pairs $((g, a), (h, b))$, and two of them, $((g, a), (h, b))$ and $((g', a'), (h', b'))$, are adjacent exactly when $a' \neq a$, $b' \neq b$, $g' \in N[g]$ and $h' \in N[h]$. Hence $a + b \pmod 2$ is constant on components, and for $c \in \{0, 1\}$ the map $((g, a), (h, a + c)) \mapsto ((g, h), a)$ is an isomorphism of the part with $a + b \equiv c$ onto $\mathsf{B}(G \boxtimes H)$, since $N[g] \times N[h] = N_{G \boxtimes H}[(g, h)]$. The value of γ^t_{gr} follows from (i) and Lemma 6.1. □

Proof of Theorem C(i)*, total part.* Let $C > 0$ and let G, H satisfy $\gamma_{\mathrm{gr}}(G \boxtimes H) > C\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$, as in Theorem 5.1. By Proposition 6.5,

$$\gamma^t_{\mathrm{gr}}(\mathsf{B}(G) \times \mathsf{B}(H)) = 4\gamma_{\mathrm{gr}}(G \boxtimes H) > 4C\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H) = C\,\gamma^t_{\mathrm{gr}}(\mathsf{B}(G))\,\gamma^t_{\mathrm{gr}}(\mathsf{B}(H)).$$

The graphs $\mathsf{B}(G)$ and $\mathsf{B}(H)$ have no isolated vertices, and they are bipartite. □

Proposition 6.5 is a concrete form of the observation of [6] that Conjecture 1.2 for bipartite graphs is equivalent to Conjecture 1.1, and the proof of (i) is the bipartite splitting used there. It turns every counterexample to Conjecture 1.1 into one to Conjecture 1.2 with the same ratio. The smallest strong-product counterexample of Theorem 5.6, $K_{2,2,2} \boxtimes H_{11}$, gives $\mathsf{B}(K_{2,2,2}) \times \mathsf{B}(H_{11})$, a graph on $12 \cdot 22 = 264$ vertices with $\gamma^t_{\mathrm{gr}} \geq 44$, while $\gamma^t_{\mathrm{gr}}(\mathsf{B}(K_{2,2,2}))\,\gamma^t_{\mathrm{gr}}(\mathsf{B}(H_{11})) = 4 \cdot 10 = 40$.

The authors of [6] checked Conjecture 1.2 by computer for all pairs of bipartite graphs with at most ten vertices, which is consistent with this size.

7. Projective planes

Let $q = p^e$ with p prime, let $\mathcal{P}$ and $\mathcal{L}$ be the sets of points and lines of $\mathrm{PG}(2,q)$, each of size q^2+q+1, and let N be the $\mathcal{P}\times\mathcal{L}$ incidence matrix, with $N_{x\ell}=1$ if $x\in\ell$ and 0 otherwise. The incidence graph $\mathrm{Inc}(q)$ has vertex set $\mathcal{P}\cup\mathcal{L}$ and an edge $x\ell$ whenever $x\in\ell$. Every point is on $q+1$ lines and every line has $q+1$ points, two points are on exactly one line and two lines meet in exactly one point. So $\mathrm{Inc}(q)$ is $(q+1)$-regular and bipartite, it has no 4-cycles, and it has girth 6; distinct vertices have distinct open neighbourhoods.

7.1. **Grundy coverings.** In a bipartite graph the terms of a total dominating sequence on one side dominate only vertices of the other side. As in the proof of Proposition 6.5, γ^t_{gr} of a bipartite graph is therefore the sum of two independent quantities. For $\mathrm{Inc}(q)$, call a sequence of points $(x_1,\dots,x_s)$ a *Grundy covering* if for each i some line through x_i misses $x_1,\dots,x_{i-1}$, and let $c(q)$ be the largest length of a Grundy covering. Dually one defines Grundy coverings of points by lines, and since $\mathrm{PG}(2,q)$ is isomorphic to its dual their largest length is again $c(q)$. Hence

$$\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(q)) = 2c(q). \tag{7.1}$$

Lemma 7.1. *For every field $\mathbb{F}$ and every matrix $M\in\mathbb{F}^{\mathcal{P}\times\mathcal{L}}$ whose nonzero entries are exactly the incident pairs, $c(q)\le \operatorname{rank}_{\mathbb{F}} M$.*

Proof. Let $(x_1,\dots,x_s)$ be a Grundy covering and let ℓ_i be a line through x_i missing $x_1,\dots,x_{i-1}$. The lines ℓ_i are distinct, since $x_j\in\ell_j$ and $x_j\notin\ell_i$ for $j<i$. The submatrix of M on rows $x_1,\dots,x_s$ and columns $\ell_1,\dots,\ell_s$ has nonzero diagonal and vanishes above the diagonal, because $M_{x_j\ell_i}=0$ for $j<i$. □

Over a field whose characteristic does not divide q the incidence matrix itself is useless here: $NN^{\mathsf T}=qI+J$ has eigenvalues $(q+1)^2$ and q, so N is nonsingular. Over $\mathbb{F}_p$ its rank is small. Hamada's formula in this case is due to Smith [15] and, independently, to MacWilliams and Mann [14]:

$$\operatorname{rank}_{\mathbb{F}_p} N = \binom{p+1}{2}^e + 1. \tag{7.2}$$

For $e=1$ this is $\binom{p+1}{2}+1=\frac12(p^2+p+2)$, about half of q^2+q+1; for $p=2$ it is 3^e+1.

Proof of Theorem D. By (7.1), Lemma 7.1 applied to N over $\mathbb{F}_p$, and (7.2),

$$\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(q)) = 2c(q) \le 2\operatorname{rank}_{\mathbb{F}_p} N = 2\Big(\tbinom{p+1}{2}^e+1\Big).$$

Since $|V(\mathrm{Inc}(q))| = 2(q^2+q+1) > 2p^{2e}$,

$$\frac{\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(p^e))}{|V(\mathrm{Inc}(p^e))|} < \Big(\frac{p+1}{2p}\Big)^e + p^{-2e} \longrightarrow 0 \qquad (e\to\infty).$$

The values for $q\in\{2,3,4,8\}$ are given in Proposition 7.2. □

The bound $c(q)\le\operatorname{rank}_{\mathbb{F}} M$ holds for every matrix with the incidence pattern, not only for N. Over $\mathbb{F}_p$ one cannot do better with another pattern matrix in the cases where the bound is attained, and in those cases the rank of the incidence matrix is the minimum rank of the pattern. For $q=5$ it is not attained.

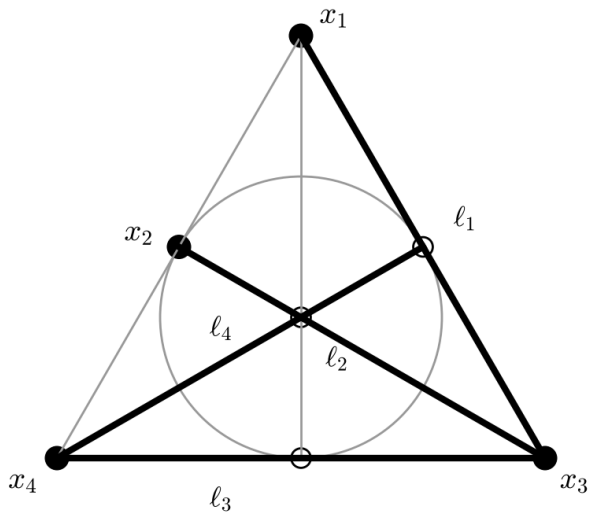


FIGURE 6. A Grundy covering of length $4 = \operatorname{rank}_{\mathbb{F}_2} N$ in the Fano plane. The points x_1, x_2, x_4 lie on a line m and x_3 lies off it; the thick lines $\ell_1, \dots, \ell_4$ satisfy $x_i \in \ell_i$ and $\ell_i \cap \{x_1, \dots, x_{i-1}\} = \emptyset$.

Proposition 7.2.

q	2	3	4	5	8
$\lvert V(\mathrm{Inc}(q))\rvert$	14	26	42	62	146
$\operatorname{rank}_{\mathbb{F}_p} N$	4	7	10	16	28
$c(q)$	4	7	10	15	28
$\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(q))$	8	14	20	30	56

Proof. The upper bounds for $q \neq 5$ are (7.2). The values of $c(q)$ for $q \leq 5$ were computed by an exhaustive search over the sets of covered lines (Appendix B), which visits 704 states for $q = 3$ and about 1.6×10^6 for $q = 5$ and also confirms the upper bound $c(5) \leq 15$. For $q = 8$ a Grundy covering of length 28 is listed in Appendix A; it was found by a randomised greedy search and is checked directly, and (7.2) shows that it is longest. □

For $q = 2$ the graph $\mathrm{Inc}(2)$ is the Heawood graph. Let m be a line of the Fano plane, x_1, x_2, x_4 its points and x_3 a point off m. Then (x_1, x_2, x_3, x_4) is a Grundy covering of length 4 (Figure 6). Indeed x_1 is on three lines; of the three lines through x_2 only m contains x_1; of those through x_3 only x_3x_1 and x_3x_2 meet $\{x_1, x_2\}$; and of those through x_4, the line m contains x_1 and x_2, the line x_4x_3 contains x_3, and the third line misses all three.

Remark 7.3. Theorem D answers the question of [10] for bipartite graphs of girth 6 and minimum degree $q + 1$, with q arbitrarily large. The graphs are regular, have no open twins and are vertex-transitive, so the obstruction is not a local one. For $p = 2$ the density $\gamma^t_{\mathrm{gr}}/|V|$ is at most $(3/4)^e$ up to a negligible error. The exact values in Proposition 7.2 suggest that $c(2^e) = 3^e + 1$ for all e (Problem 9.4).

8. Computations

None of the proofs depends on a computer. The computations below were used to find the constructions and to test the statements, and they are reproducible from the repository named in the data availability statement.

TABLE 3. The octahedron times H_N. The row $2\gamma_{\mathrm{gr}}(H_N)$ is the value predicted by Conjecture 1.1, and the last row the exact value, which equals $\max(N, 2\gamma_{\mathrm{gr}}(H_N))$ in this range; bold entries are counterexamples.

N	4	5	6	7	8	9	10	11	12	13	14	15
$\gamma_{\mathrm{gr}}(H_N)$	3	3	4	4	4	5	5	5	6	6	6	7
$2\gamma_{\mathrm{gr}}(H_N)$	6	6	8	8	8	10	10	10	12	12	12	14
$\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N)$	6	6	8	8	8	10	10	**11**	12	**13**	**14**	**15**

8.1. **An exact solver.** A dominating sequence is determined, as far as its continuations are concerned, by the set D of vertices it has dominated so far: a vertex v can be appended exactly when $N[v] \not\subseteq D$, and the new set is $D \cup N[v]$. So $\gamma_{\mathrm{gr}}(G) = f(\emptyset)$ for the function on subsets

$$f(D) = \max\big(\{0\} \cup \{1 + f(D \cup N[v]) : N[v] \not\subseteq D\}\big),$$

and the same recursion with open neighbourhoods gives γ^t_{gr}. We evaluate f by depth-first search with a hash table keyed by D, stored as a 128-bit mask, so graphs with up to 128 vertices can be handled; the cost is the number of reachable sets D. Two simple reductions make the search feasible for the graphs of this paper: closed twins are merged (Remark 2.6), and for bipartite graphs and γ^t_{gr} the two sides are treated separately, which is (7.1). The values obtained are:

- $\gamma_{\mathrm{gr}}(C_k \boxtimes C_\ell) = (k-2)(\ell-2)$ for $3 \le k \le \ell \le 7$, and $\gamma_{\mathrm{gr}}(P \boxtimes C_5) = 15$, where P is the Petersen graph and $\gamma_{\mathrm{gr}}(P) = 5$;
- γ^t_{gr} of the direct products in Table 2, $\gamma^t_{\mathrm{gr}}(C_n) = 2\lfloor (n-1)/2 \rfloor$ for $3 \le n \le 14$, and $\gamma^t_{\mathrm{gr}}(T) = 2\nu(T)$ for thirty random trees with up to sixteen vertices;
- $\gamma_{\mathrm{gr}}(H_N) = \lfloor N/3 \rfloor + 2$ for $4 \le N \le 24$, and the values of $\gamma_{\mathrm{gr}}(K_{2,2,2} \boxtimes H_N)$ in Table 3;
- $c(q)$ for $q \in \{2,3,4,5\}$ in Proposition 7.2.

8.2. **Ranks.** The ranks used in Sections 3, 5 and 7 were recomputed independently: by Gaussian elimination over the prime fields for $A_0 + I$ of $K(n,2)$ over $\mathbb{F}_2$ ($n = 6, 8, \dots, 14$), for $A_0 - I$ over $\mathbb{F}_p$ for the pairs $(n,p) = (7,3), (9,5), (10,3), (11,7), (13,3), (14,5)$, for the matrices of Theorem 5.1 with $7 \le m \le 10$, and for the p-ranks (7.2) with $q \in \{2,3,4,5,7,8\}$; and by computing eigenvalues for the circulants $A - 2\cos(2\pi/k)I$ of C_k and the matrices A_ω of Proposition 6.3, for $3 \le k, n \le 30$. The factorizations behind the rank bounds for $K(n,2)$ with $n \in \{6,7,8,9,10\}$ and for the Fano plane are certified in Lean (Appendix B).

8.3. **A search for small counterexamples.** Let $\mathcal{G}$ be the set of connected graphs with 4 to 7 vertices that are not forests and have no closed twins; forests are excluded because Conjecture 1.1 holds when one factor is a forest [2], and closed twins because they do not change the problem (Remark 2.6). There are 560 such graphs. For each of the 157,080 unordered pairs $\{G, H\} \subseteq \mathcal{G}$ we computed $\gamma_{\mathrm{gr}}(G)$ and $\gamma_{\mathrm{gr}}(H)$ exactly and ran a randomised greedy search for a dominating sequence of $G \boxtimes H$ longer than $\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$. None was found. This is not a proof that the conjecture holds for these pairs, but together with Theorem B it suggests that counterexamples need a factor with a large gap between γ_{gr} and mr, as H_N has, and are not small. The smallest counterexample we know, $K_{2,2,2} \boxtimes H_{11}$, has 66 vertices.

9. STATUS OF THE CONJECTURES AND OPEN PROBLEMS

Table 4 records the status of every conjecture and question on Grundy domination of products and of bipartite graphs treated in this paper.

TABLE 4. Conjectures and questions treated in this paper.

statement	status	where
$\gamma_{\mathrm{gr}}(G \boxtimes H) = \gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$ for all G, H [7, Conj. 22]	false, with unbounded ratio	Theorems 5.1, 5.6
$\gamma_{\mathrm{gr}}(C_k \boxtimes C_\ell) = (k-2)(\ell-2)$ (conjectural value in [7])	true	Theorem B
$\gamma^t_{\mathrm{gr}}(G \times H) = \gamma^t_{\mathrm{gr}}(G)\gamma^t_{\mathrm{gr}}(H)$ [6, Conj. 2.2]	false, with unbounded ratio	Proposition 6.5
$\gamma^t_{\mathrm{gr}}(C_k \times C_\ell) = \gamma^t_{\mathrm{gr}}(C_k)\gamma^t_{\mathrm{gr}}(C_\ell)$ [6]	extended to any number of cycles, forests, complete and complete bipartite graphs	Theorem 6.4
Is there $c > 0$ with $\gamma^t_{\mathrm{gr}}(G) \geq c\lvert V(G)\rvert$ for bipartite C_4-free G with $\delta(G) \geq 2$? [10, Question 1]	no	Theorem D
$\gamma_{\mathrm{gr}}(K(n,2)) = n-1$ for $n \geq 6$ [12]	new proof of the upper bound	Proposition 3.8

Problem 9.1. *Find the smallest number of vertices of a strong product $G \boxtimes H$ with $\gamma_{\mathrm{gr}}(G \boxtimes H) > \gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H)$. We know that it is at most* 66*, and Section 8 suggests that it exceeds* 49*.*

Problem 9.2. *Characterise the graphs that are $\mathbb{F}$-tight for some field $\mathbb{F}$. In particular, is every tree tight over some field? A positive answer would give a new proof of the theorem of Bell, Driscoll, Krop and Wolff* [2] *restricted to tight second factors. Paths and stars are tight over $\mathbb{R}$ (Propositions 3.3 and 3.4).*

Problem 9.3. *How fast can $\gamma_{\mathrm{gr}}(G \boxtimes H)/(\gamma_{\mathrm{gr}}(G)\gamma_{\mathrm{gr}}(H))$ grow as a function of $\lvert V(G)\rvert + \lvert V(H)\rvert$? Theorem 5.1 gives growth of order $n^{2/5}$ in terms of the number $n = \binom{m}{5}$ of vertices of each factor. The trivial upper bound is $\gamma_{\mathrm{gr}}(G \boxtimes H) \leq \lvert V(G)\rvert\lvert V(H)\rvert$.*

Problem 9.4. *Is $c(2^e) = 3^e + 1$ for every e, so that $\gamma^t_{\mathrm{gr}}(\mathrm{Inc}(2^e)) = 2(3^e+1)$? What is $c(p)$ for odd primes p? We have $c(5) = 15$, one less than the* 5*-rank, and for $q = 7$ a randomised search found a Grundy covering of length* 25*, while the* 7*-rank is* 29*.*

Problem 9.5. *For a graph G let $g(G) = \min_{\mathbb{F}} \mathrm{mr}_{\mathbb{F}}(G) - \gamma_{\mathrm{gr}}(G)$, the minimum over all fields. Corollary 5.7 gives $g(H_N) \geq N/2 - \lfloor N/3 \rfloor - 2$. Is there an $\epsilon > 0$ and graphs of every order with $g(G) \geq \epsilon\lvert V(G)\rvert$? Large gaps are what counterexamples to Conjecture 1.1 are made of, by Proposition 4.4.*

APPENDIX A. EXPLICIT SEQUENCES

The Petersen graph. With the vertices of the Petersen graph written as the two-element subsets of $\{1, \ldots, 5\}$, adjacent when disjoint, the sequence $\{1,2\}, \{1,3\}, \{2,4\}, \{1,5\}, \{1,4\}$ is a dominating sequence with footprints $\{1,2\}, \{1,3\}, \{1,5\}, \{2,3\}, \{1,4\}$, so $\gamma_{\mathrm{gr}} \geq 5$; Proposition 3.8 gives equality.

The octahedron times H_{11}. In the notation of Section 5, the eleven vertices $(g_{-j \bmod 3}, j)$, $0 \leq j \leq 10$, in increasing order of j, form a dominating sequence of $K_{2,2,2} \boxtimes H_{11}$. Footprints are $(g_0, 0)$, $(g_0, 2)$, $(g_0, 3)$ for $j = 0, 1, 2$ and (x_{1-j}, j) for $j \geq 3$; the reader can check this from (5.2) and the fact that x_t is adjacent to every vertex of the octahedron except g_t. The sequence is certified in Lean.

A Grundy covering of $\mathrm{PG}(2,8)$**.** Write $\mathbb{F}_8 = \mathbb{F}_2[x]/(x^3+x+1)$ and represent $a_0+a_1x+a_2x^2$ by the integer $a_0+2a_1+4a_2$. A point (p_0,p_1,p_2) lies on the line $[l_0,l_1,l_2]$ when $p_0l_0+p_1l_1+p_2l_2=0$. Table 5 lists a Grundy covering $(x_1,\dots,x_{28})$ of length 28 together with lines $\ell_i \ni x_i$ that miss $x_1,\dots,x_{i-1}$.

TABLE 5. A Grundy covering of $\mathrm{PG}(2,8)$ of length $28 = 3^3+1$.

i	x_i	ℓ_i	i	x_i	ℓ_i
1	$(1,2,3)$	$[0,1,7]$	2	$(1,0,1)$	$[0,1,0]$
3	$(1,5,7)$	$[0,1,2]$	4	$(1,3,7)$	$[1,1,3]$
5	$(1,5,1)$	$[0,1,5]$	6	$(1,0,2)$	$[1,0,5]$
7	$(1,4,3)$	$[1,2,7]$	8	$(1,4,4)$	$[0,1,1]$
9	$(1,3,4)$	$[1,7,2]$	10	$(1,3,0)$	$[0,0,1]$
11	$(1,6,7)$	$[1,3,0]$	12	$(1,5,5)$	$[1,0,2]$
13	$(1,2,0)$	$[1,5,4]$	14	$(1,6,3)$	$[1,2,2]$
15	$(1,1,5)$	$[1,1,0]$	16	$(1,1,2)$	$[1,2,4]$
17	$(1,1,1)$	$[1,7,6]$	18	$(1,3,5)$	$[0,1,6]$
19	$(1,0,7)$	$[1,7,4]$	20	$(1,2,6)$	$[1,0,3]$
21	$(1,7,1)$	$[1,4,0]$	22	$(1,7,5)$	$[1,1,7]$
23	$(1,3,6)$	$[1,3,7]$	24	$(1,2,1)$	$[1,4,2]$
25	$(1,0,5)$	$[1,1,2]$	26	$(0,0,1)$	$[1,0,0]$
27	$(1,7,3)$	$[0,1,4]$	28	$(0,1,6)$	$[0,1,3]$

APPENDIX B. MACHINE CHECKS

Three independent programs were used; none of the proofs relies on them.

The exact solver. A C program evaluates the recursion of Section 8 with 128-bit masks and a hash table. A Python driver produces the graphs and runs the randomised greedy search of Section 8.

Julia. A separate Julia program, written without reference to the C code, recomputes with its own memoised recursion the values of γ_{gr} for $C_k \boxtimes C_\ell$ $(3 \le k \le \ell \le 6)$, the Petersen graph and its product with C_5, H_N $(4 \le N \le 20)$, $K_{2,2,2} \boxtimes H_N$ $(4 \le N \le 12)$ and $K(n,2)$ $(6 \le n \le 12)$; the values of γ_{gr}^t for C_n $(3 \le n \le 14)$ and for six direct products of cycles; the support patterns and ranks of the matrices of Theorem 5.1 for $7 \le m \le 10$; the ranks of Section 8; and $c(q)$ for $q \in \{2,3,5\}$. All 153 checks pass.

Lean. A Lean 4 file using only the core library certifies, by kernel evaluation (`decide`), the dominating sequences above, a dominating sequence of length 5 of H_{11}, the Grundy coverings of the Fano plane and of $\mathrm{PG}(2,8)$, and the factorizations $M = CR$ over $\mathbb{F}_p$, with C having $n-1$ columns, for $A_0 + I$ of $K(n,2)$ over $\mathbb{F}_2$ $(n = 6,8,10)$ and $A_0 - I$ over $\mathbb{F}_3$ $(n = 7,10)$ and $\mathbb{F}_5$ $(n = 9)$, and with 4 columns for the incidence matrix of the Fano plane. The file has 13 theorems; twelve use no axioms and one uses only `propext`.

AUTHOR CONTRIBUTIONS

Deep Bhattacharjee is the sole author. He posed the problems in the form treated here, proved the results, wrote the paper, drew the figures, and wrote the programs and the Lean certificate.

Open Gluon and Anaxia

Two tools built by the author organised this project. Open Gluon is a workspace that keeps pinned copies of the mathematical libraries, the source record of the project and its checks in one reproducible place. Anaxia, the engine of the eBLLM project, stores each construction as a route with its input, its output, the obligations that must be discharged and a test that makes the route fail when an obligation is skipped, and it certifies finite statements in exact integer arithmetic. It was used here to keep the sequences, matrices and their checks in one record. Every statement of this paper is proved in the text.

Declarations

Data and code availability. No data were generated. The exact solver, the Julia programs and the Lean 4 certificate are available at `https://github.com/creelie/Paper_1` and archived on Zenodo under the DOI 10.5281/zenodo.23607651.

Funding. No funding of any kind was received for this work.

Competing interests. The author declares no competing interests.

Ethics approval. Not applicable. The work involves no human participants, animals or personal data.

References

[1] N. Alon, *The Shannon capacity of a union*, Combinatorica **18** (1998), no. 3, 301–310, doi:10.1007/PL00009824.

[2] K. Bell, K. Driscoll, E. Krop and K. Wolff, *Grundy domination of forests and the strong product conjecture*, Electron. J. Combin. **28** (2021), no. 2, Paper No. P2.12, doi:10.37236/9507.

[3] D. Bhattacharjee, *Establishing equivariant class* $[O]$ *for hyperbolic groups*, Asian Res. J. Math. **18** (2022), no. 11, 362–369, doi:10.9734/arjom/2022/v18i11615.

[4] D. Bhattacharjee, *Non-commutativity over canonical suspension* η *for genus* $g \geq 1$ *in hypercomplex structures for potential* ρ_ϕ, Asian Res. J. Math. **18** (2022), no. 11, 332–341, doi:10.9734/arjom/2022/v18i11607.

[5] D. Bhattacharjee and U. Bhattacharya, *Iterated nerve complexes for k-fold sensor coverage*, Preprints.org (2026), doi:10.20944/preprints202606.1351.v2.

[6] B. Brešar, Cs. Bujtás, T. Gologranc, S. Klavžar, G. Košmrlj, T. Marc, B. Patkós, Zs. Tuza and M. Vizer, *On Grundy total domination number in product graphs*, Discuss. Math. Graph Theory **41** (2021), no. 1, 225–247, doi:10.7151/dmgt.2184.

[7] B. Brešar, Cs. Bujtás, T. Gologranc, S. Klavžar, G. Košmrlj, B. Patkós, Zs. Tuza and M. Vizer, *Dominating sequences in grid-like and toroidal graphs*, Electron. J. Combin. **23** (2016), no. 4, Paper No. P4.34, doi:10.37236/6269.

[8] B. Brešar, Cs. Bujtás, T. Gologranc, S. Klavžar, G. Košmrlj, B. Patkós, Zs. Tuza and M. Vizer, *Grundy dominating sequences and zero forcing sets*, Discrete Optim. **26** (2017), 66–77, doi:10.1016/j.disopt.2017.07.001.

[9] B. Brešar, T. Gologranc, M. Milanič, D. F. Rall and R. Rizzi, *Dominating sequences in graphs*, Discrete Math. **336** (2014), 22–36, doi:10.1016/j.disc.2014.07.016.

[10] B. Brešar, M. A. Henning and D. F. Rall, *Total dominating sequences in graphs*, Discrete Math. **339** (2016), no. 6, 1665–1676, doi:10.1016/j.disc.2016.01.017.

[11] B. Brešar, T. Kos, G. Nasini and P. Torres, *Total dominating sequences in trees, split graphs, and under modular decomposition*, Discrete Optim. **28** (2018), 16–30, doi:10.1016/j.disopt.2017.10.002.

[12] B. Brešar, T. Kos and P. Torres, *Grundy domination and zero forcing in Kneser graphs*, Ars Math. Contemp. **17** (2019), no. 2, 419–430, doi:10.26493/1855-3974.1881.384.

[13] J. C.-H. Lin, *Zero forcing number, Grundy domination number, and their variants*, Linear Algebra Appl. **563** (2019), 240–254, doi:10.1016/j.laa.2018.11.003.

[14] F. J. MacWilliams and H. B. Mann, *On the p-rank of the design matrix of a difference set*, Inform. Control **12** (1968), no. 5, 474–488, doi:10.1016/S0019-9958(68)90534-2.

[15] K. J. C. Smith, *On the p-rank of the incidence matrix of points and hyperplanes in a finite projective geometry*, J. Combin. Theory **7** (1969), no. 2, 122–129, doi:10.1016/S0021-9800(69)80046-3.

Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India

Formerly, University of Calcutta (Heramba Chandra College), 23/49 Gariahat Road, Golpark, Kolkata 700029, West Bengal, India
Email address: d.bhattacharjee@erl-forschung.de
Email address: itsdeep@live.com
Email address: egspl@proton.me (GitHub-related contact; GitHub: creelie)
URL: https://orcid.org/0000-0003-0466-750X